MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 3

The Formation of the Solar System and the Hadean Earth and its Environment

4567-4031 Ma

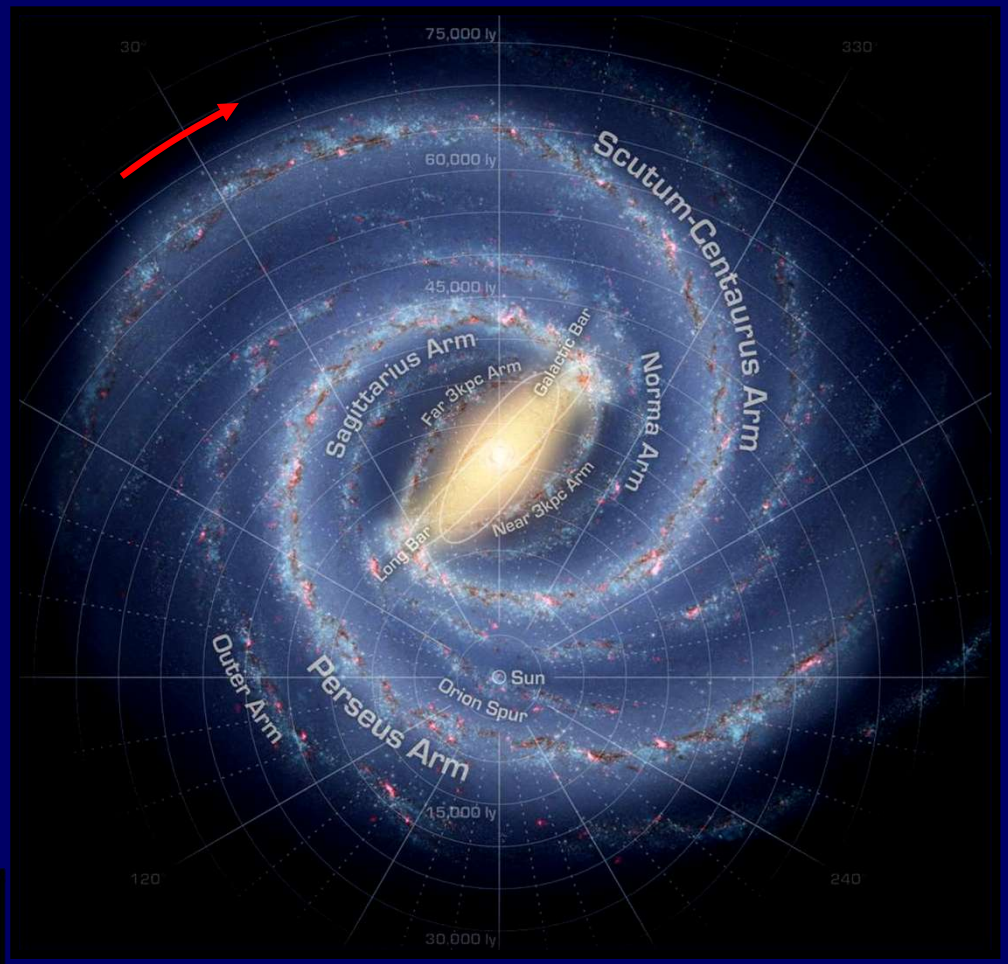
age of the solar system

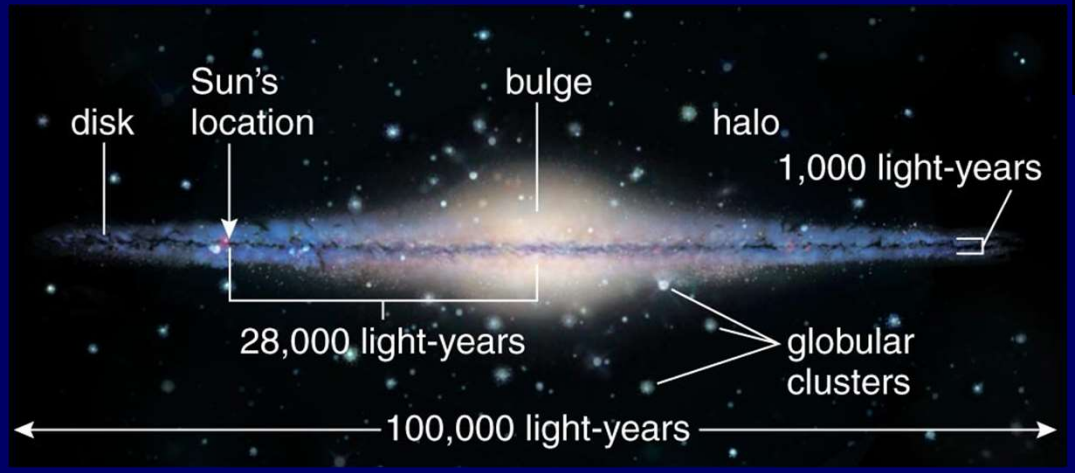
oldest rocks



Milky Way galaxy- geography and geometry recap

- a rotating, barred spiral galaxy
- disc of stars ~ 100,000 light years across by ~ 1000 light years thick
- bulges near the centre
- surrounded by spherical halo containing globular star clusters
- has a supermassive black hole (Sagittarius A*) at its centre

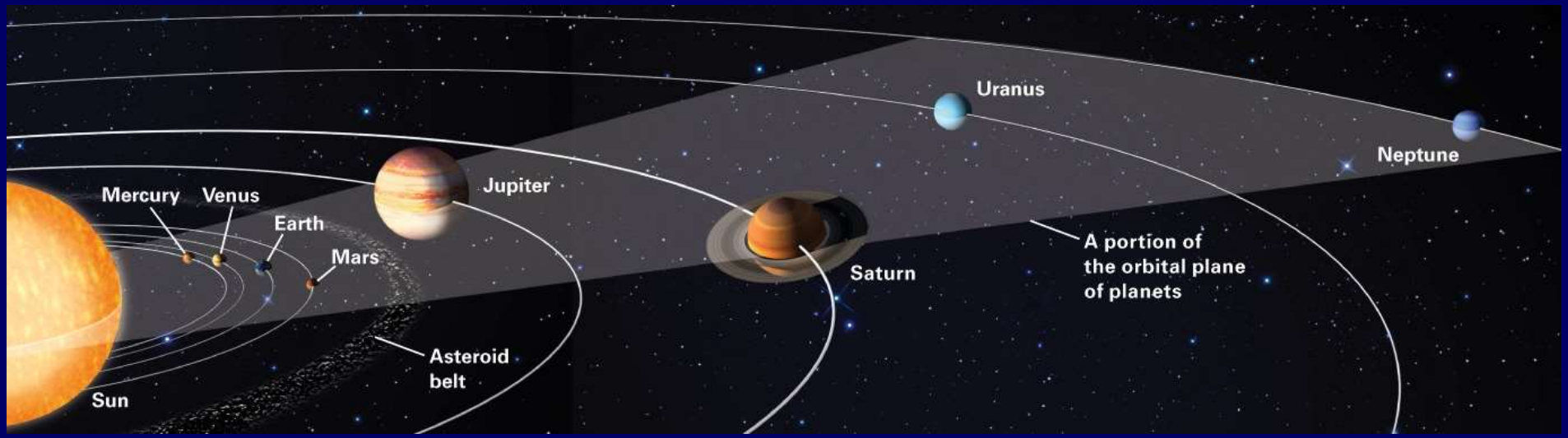




• our Sun is a Population I star located in the galactic disc ~28,000 light years from the galactic centre, just above the galactic mid-plane

Overview of our Solar System

4567 Ma



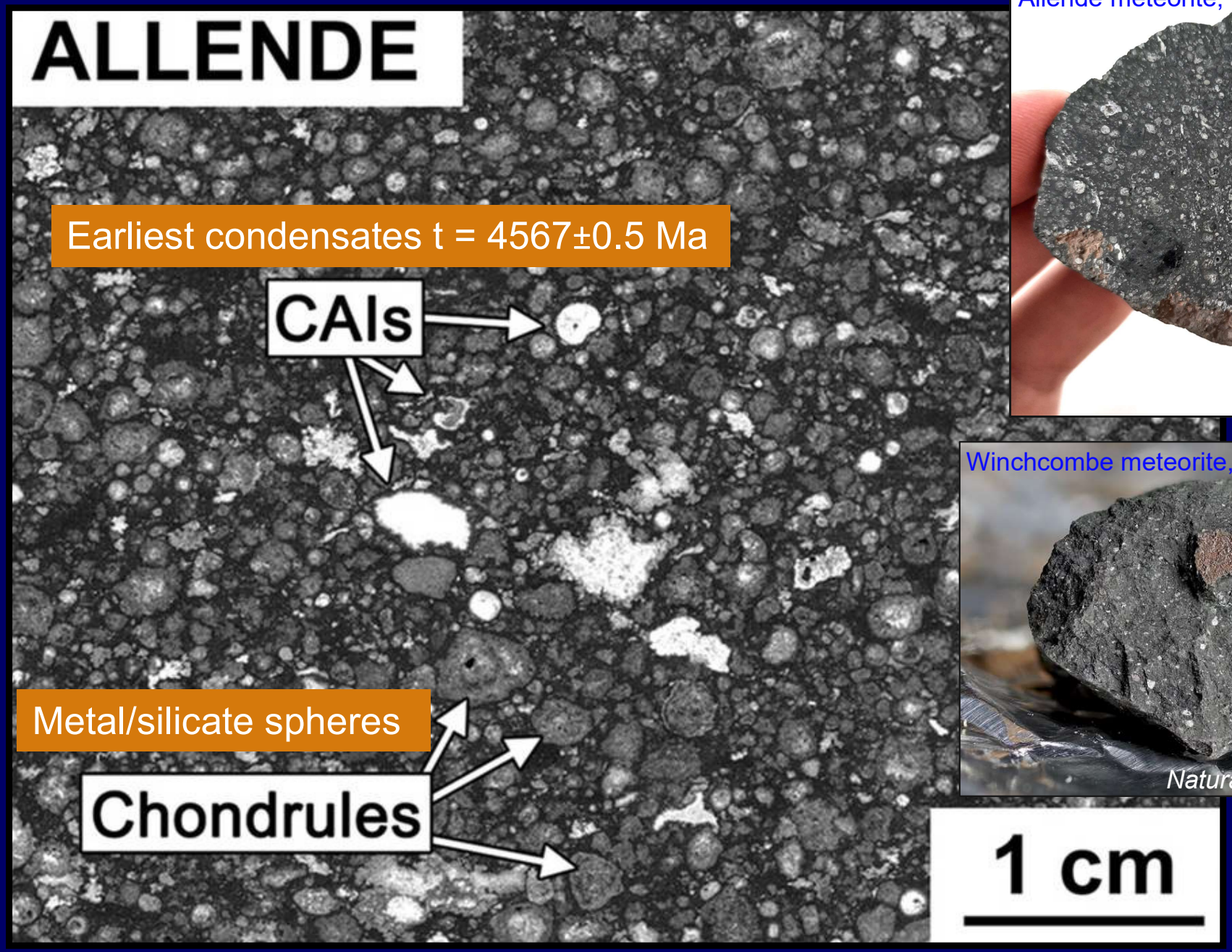
1 Star, 8 planets (4 rocky, 4 gas giants), several dwarf planets (Pluto) and an asteroid belt

Lecture highlights



- the formation of planets from a solar nebular and the timescales involved in these processes
- the age of our own solar system and the evidence for this
- the processes and time scales over which the Earth's interior became gravitationally differentiated
- when the Moon formed and compare hypotheses for its origin
- key differences between the inner and outer planets of the solar system
- comparison of the early Earth in the Hadean to the present day in terms of tectonic style and environmental conditions

Meteorites: a Window on the Soul of the Solar System







Forming the Solar System: Nebula Hypothesis

- Sun and planets formed by gravitational collapse of an interstellar gas and dust (a molecular cloud) at about same time
- ➢ Sun and planets about the same age

- Solar nebula: huge volume of thinly dispersed gas and dust

- Gas is mainly hydrogen and helium

- Dust is fine-grained 0.1 micron – mineral grains (contain heavier elements of silica and iron)

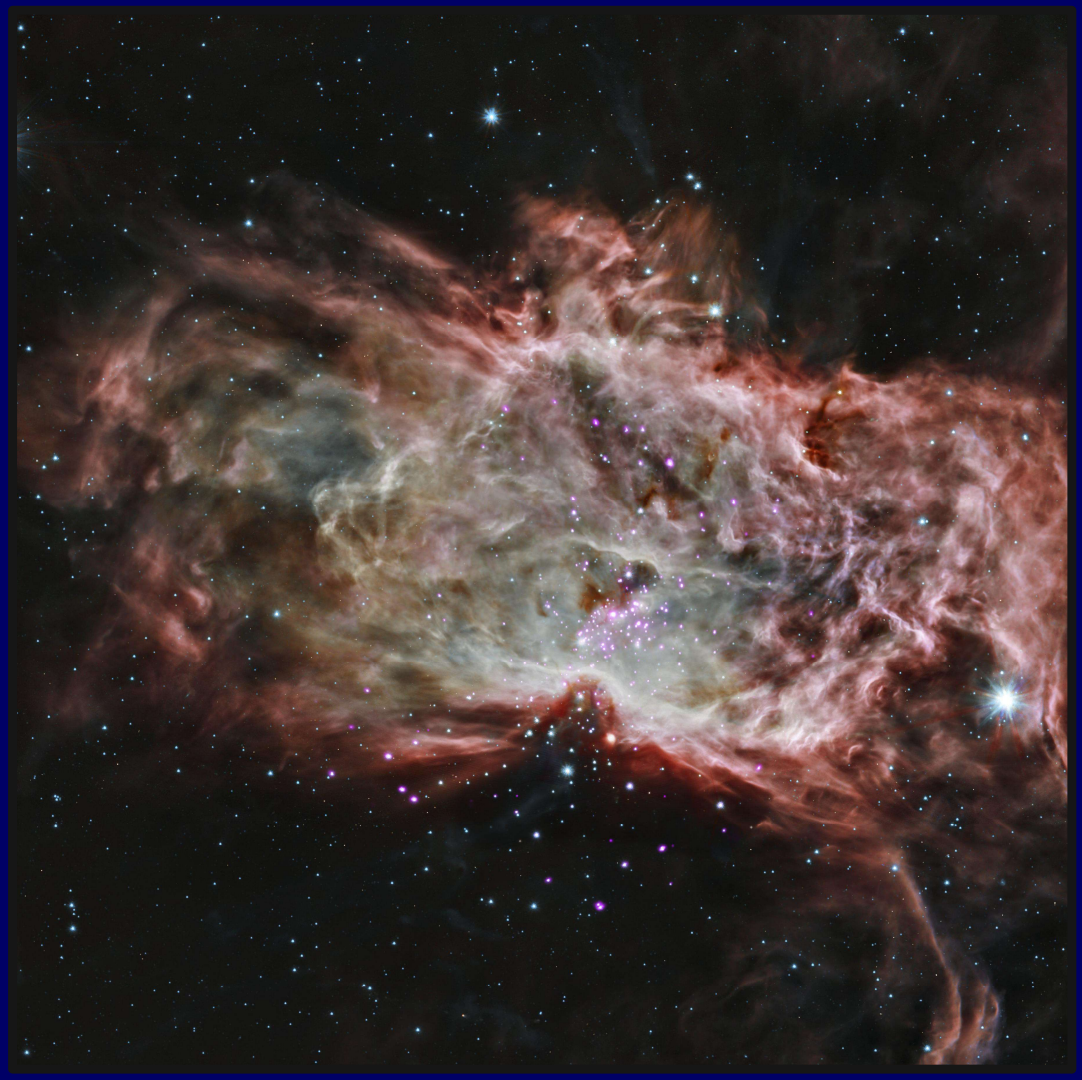
- Temperature 10-50K

- Within this cold, dense, magnetized cloud lumps of higher density form due to turbulent motions



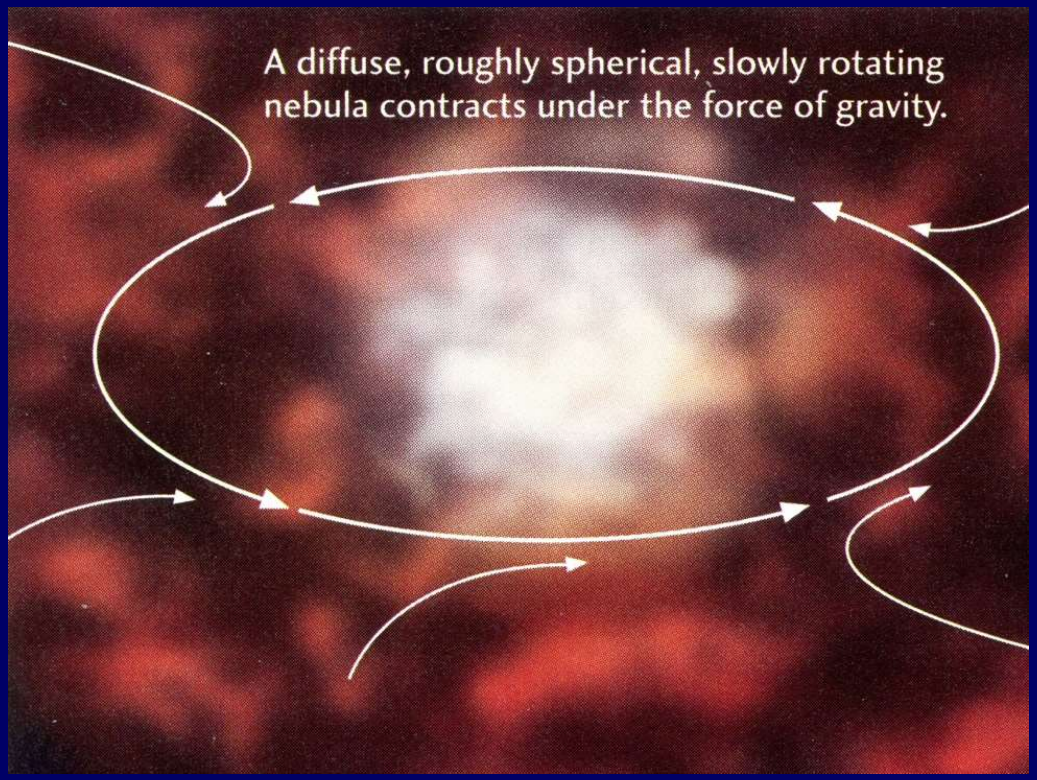
Gravitational collapse and contraction of the Solar nebula

- A nearby supernova shock wave triggered gravitational collapse and fragmentation of the cloud
- The fragments themselves collapsed and fragmented into smaller and smaller pieces
- As individual fragments continued to collapse, energy released by the collapse was radiated away and so temperature did not rise much
- Initial molecular cloud: 10-100 light years across, but fragment < 1 light year across and < 2 solar masses
- A fragment of the size required to form a Sun-sized star took a few 10 kyr to collapse to the size of the Solar System
- At this point, the core became too dense for the radiation to escape and the temperature started to rise



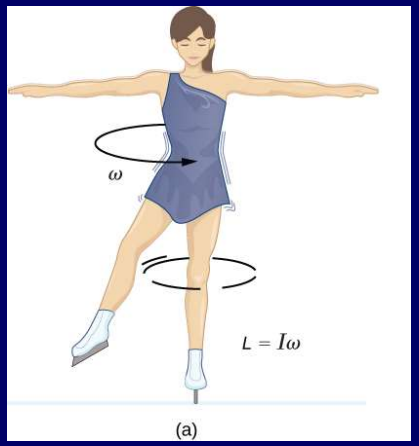
Flame nebula

The evolving solar nebula



Spherical cloud:

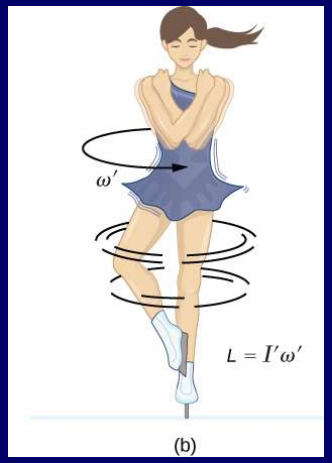
- diameter 1 light year (9.4607 x 10^{12} km)
- rotates once every 10 myr (10^{-7} rev/yr)

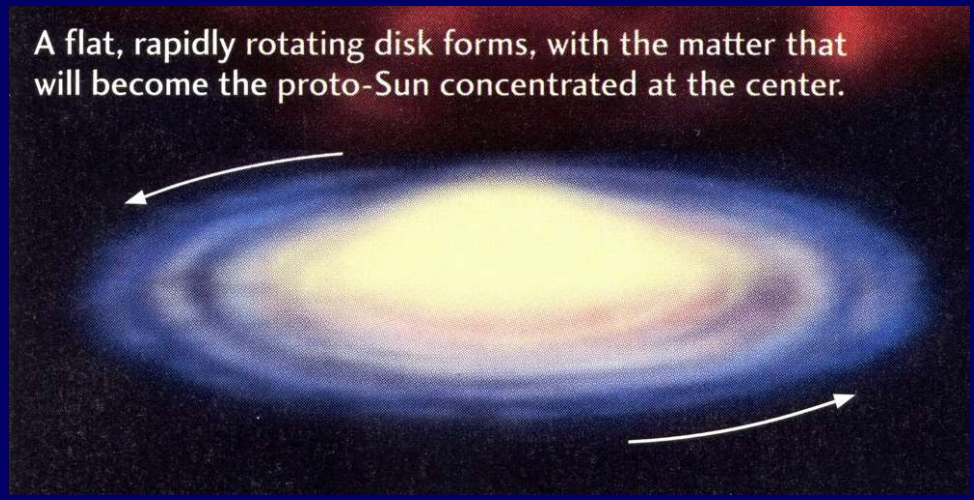


Angular momentum =
mass x rotation-rate x radius^2

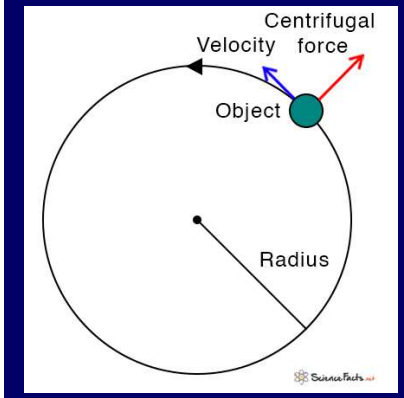
Cloud shrunk to diameter of orbit of Neptune

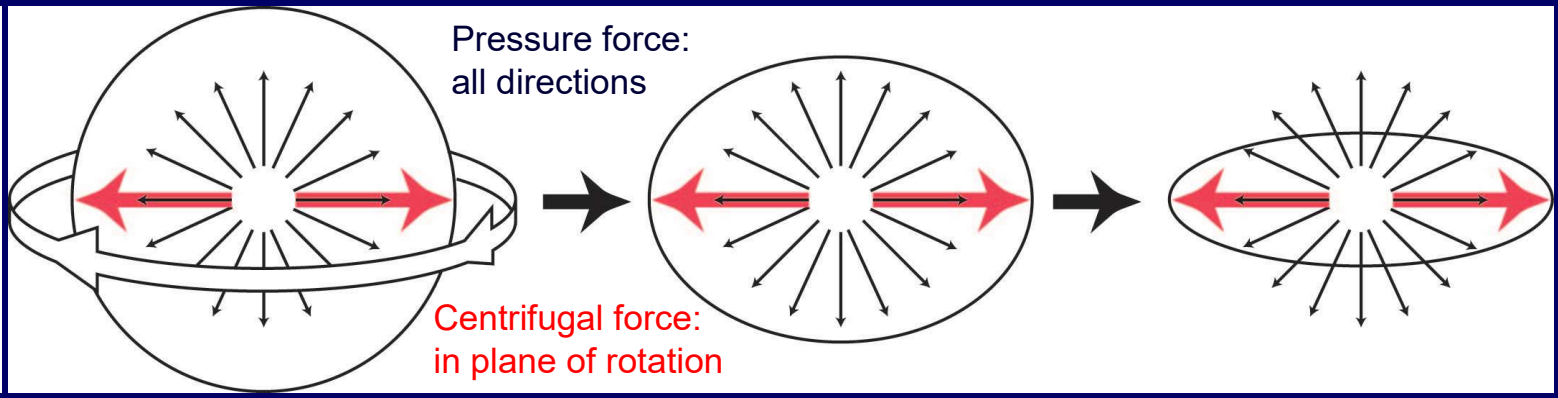
- same mass
- new diameter 9 x 10^9 km
- rotates once every 9 yr (0.1105 rev/yr)





Formation of a protoplanetary disc





Centrifugal force:
plane of rotation

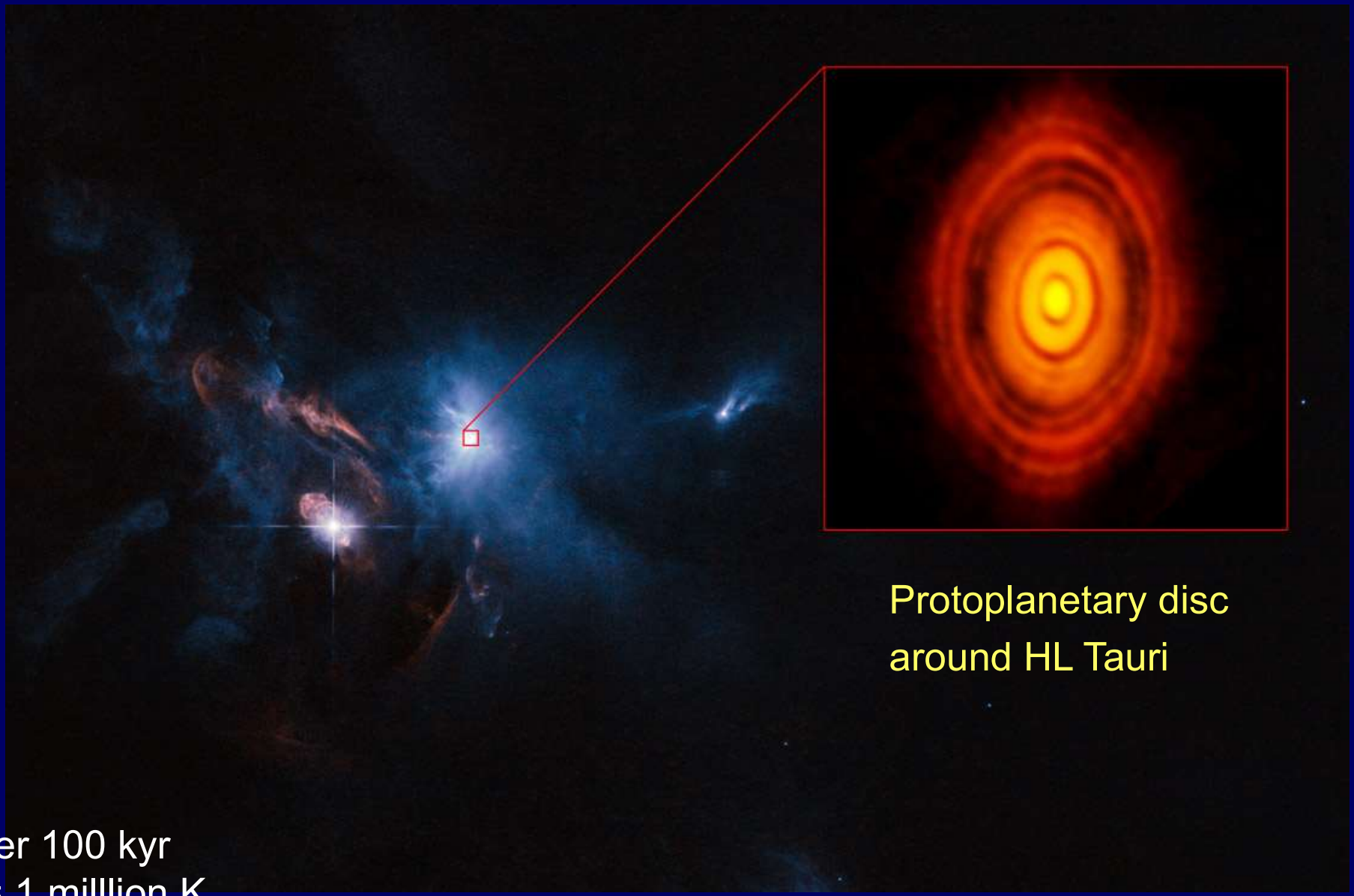
The formation of the protoplanetary disc





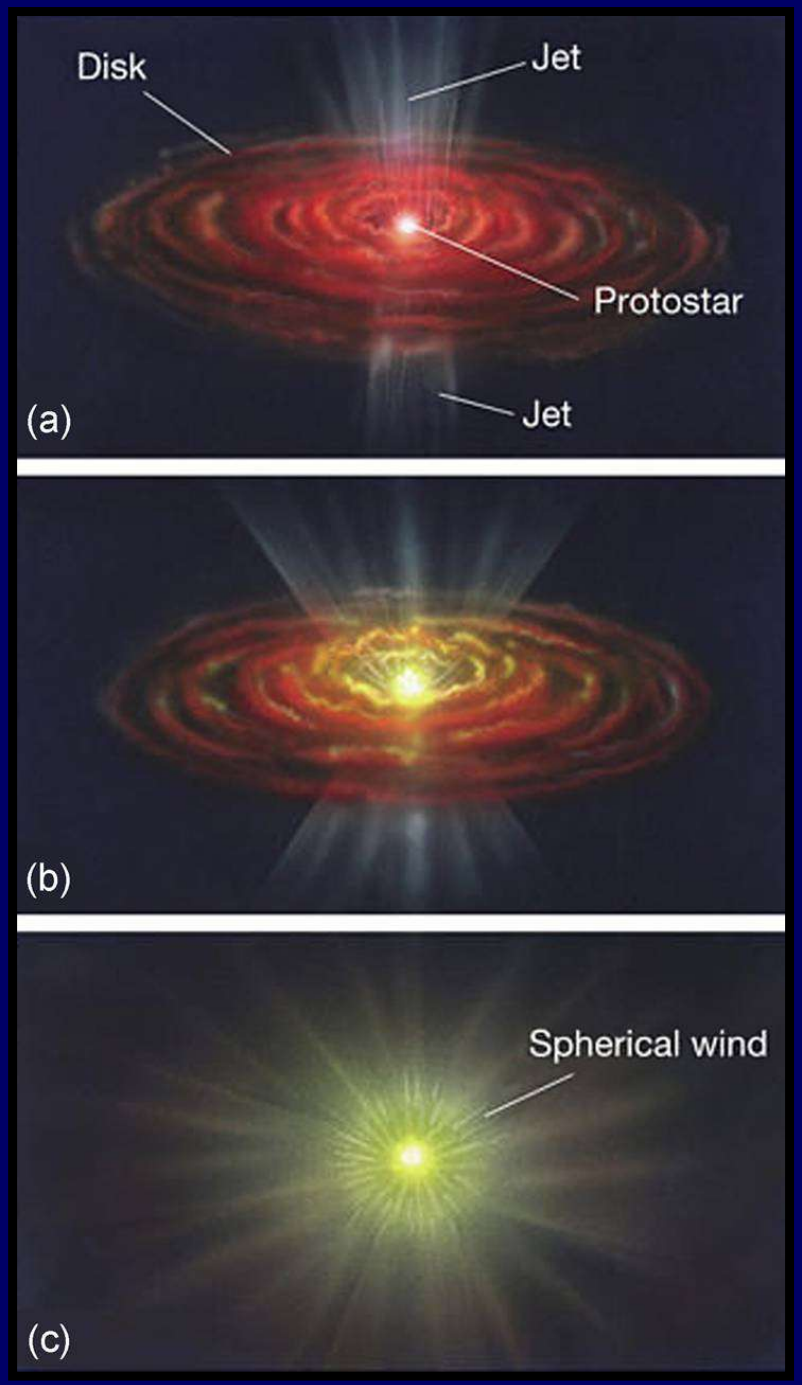
The Rotor Ride- example of centrifugal force in a rotating frame of reference

The evolving solar nebula

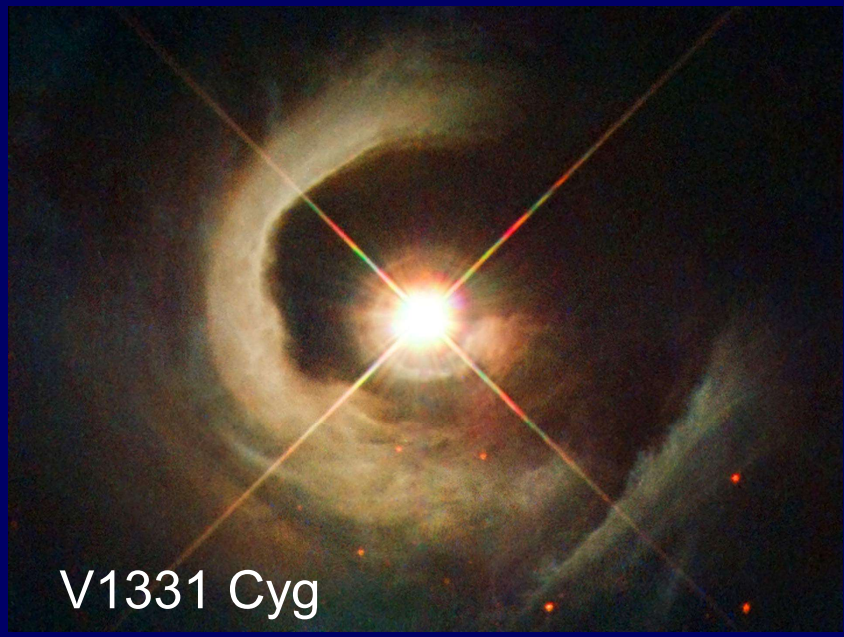


After 100 kyr
T = 1 milllion K
Protostar formation: T-Tauri phase

Solar wind, bipolar jets and disc development

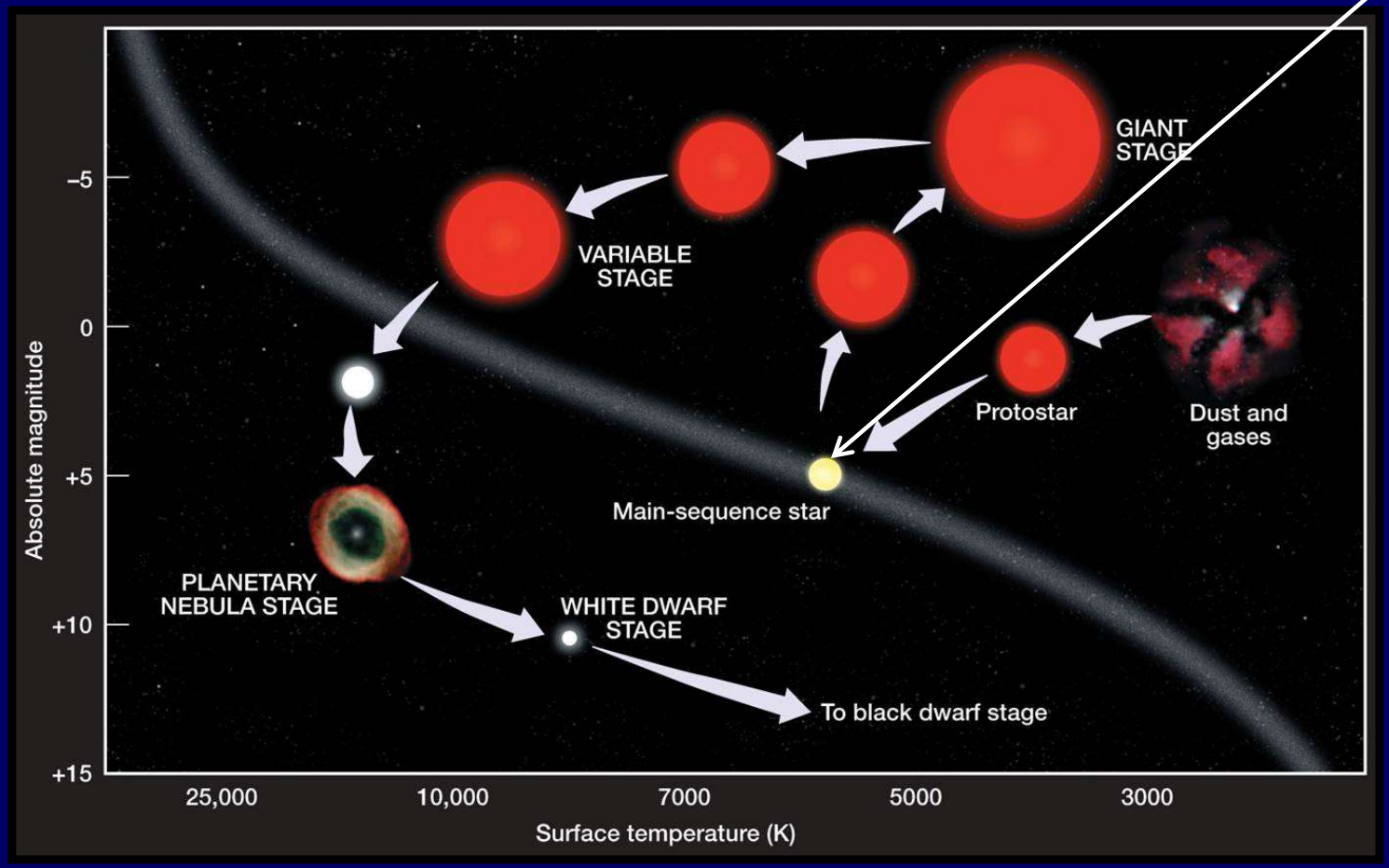


- T Tauri protostars are characterized by strong winds driven by heat of gravitational contraction, not fusion
- a bipolar outflow expel two jets of matter out in directions perpendicular to the protoplanetary disc
- inner part of disc cleared of finer dust and gas and proto-Sun become visible
- widening of jets to become spherical wind
- in 10,000 yr jets and winds expel ½ a solar mass, clear inner Solar system of orbiting bodies <10 m in size and gas, helping to redistribute mass outwards



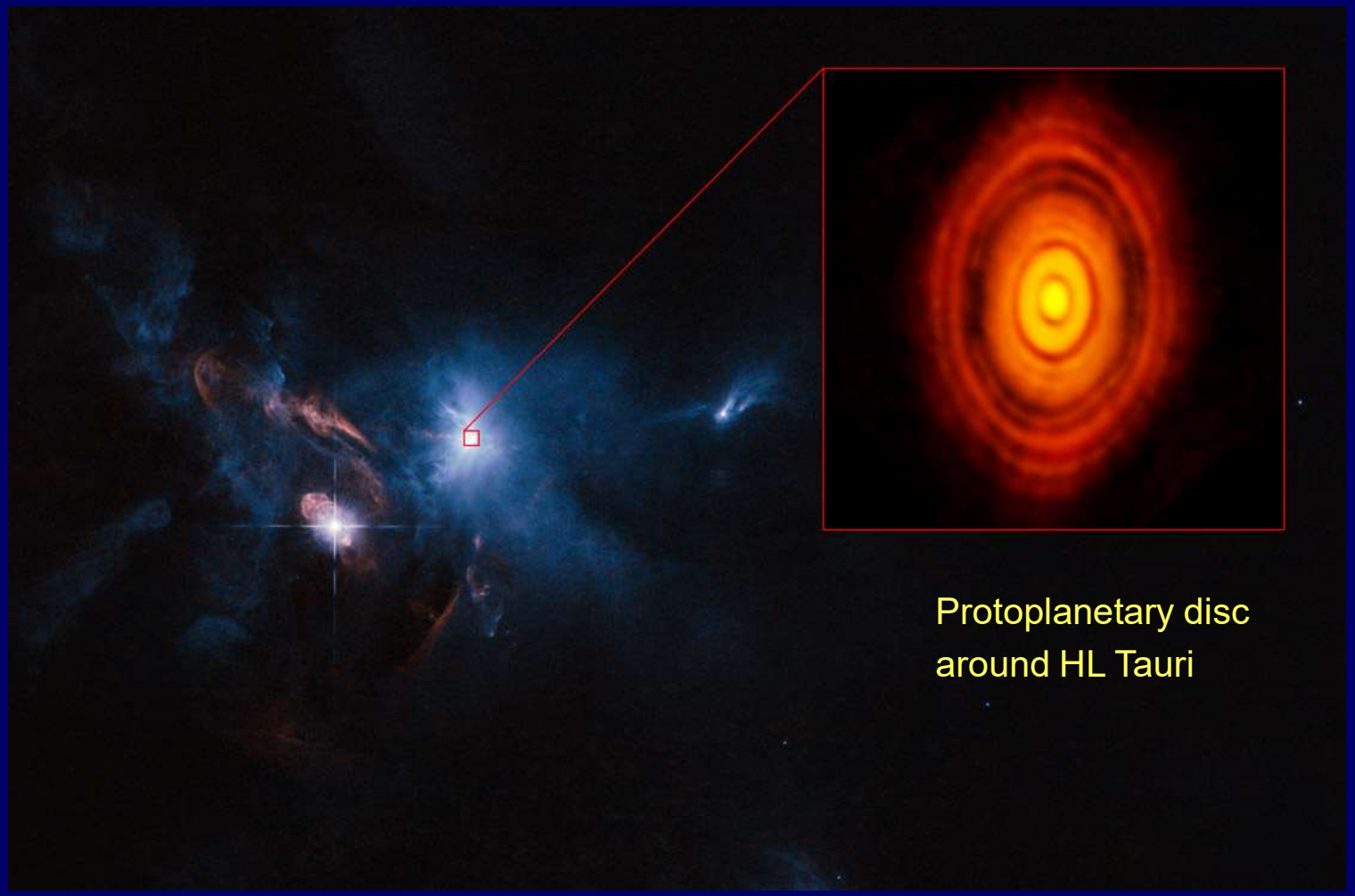
The Sun becomes a main sequence star

Equilibrium state between expansion and contraction = main sequence



After 10 Ma
T in core raised to 10 million K
Hydrogen fusion reactions initiated, leading to end of gravitational collapse
Protostar → true star (Main-sequence star)

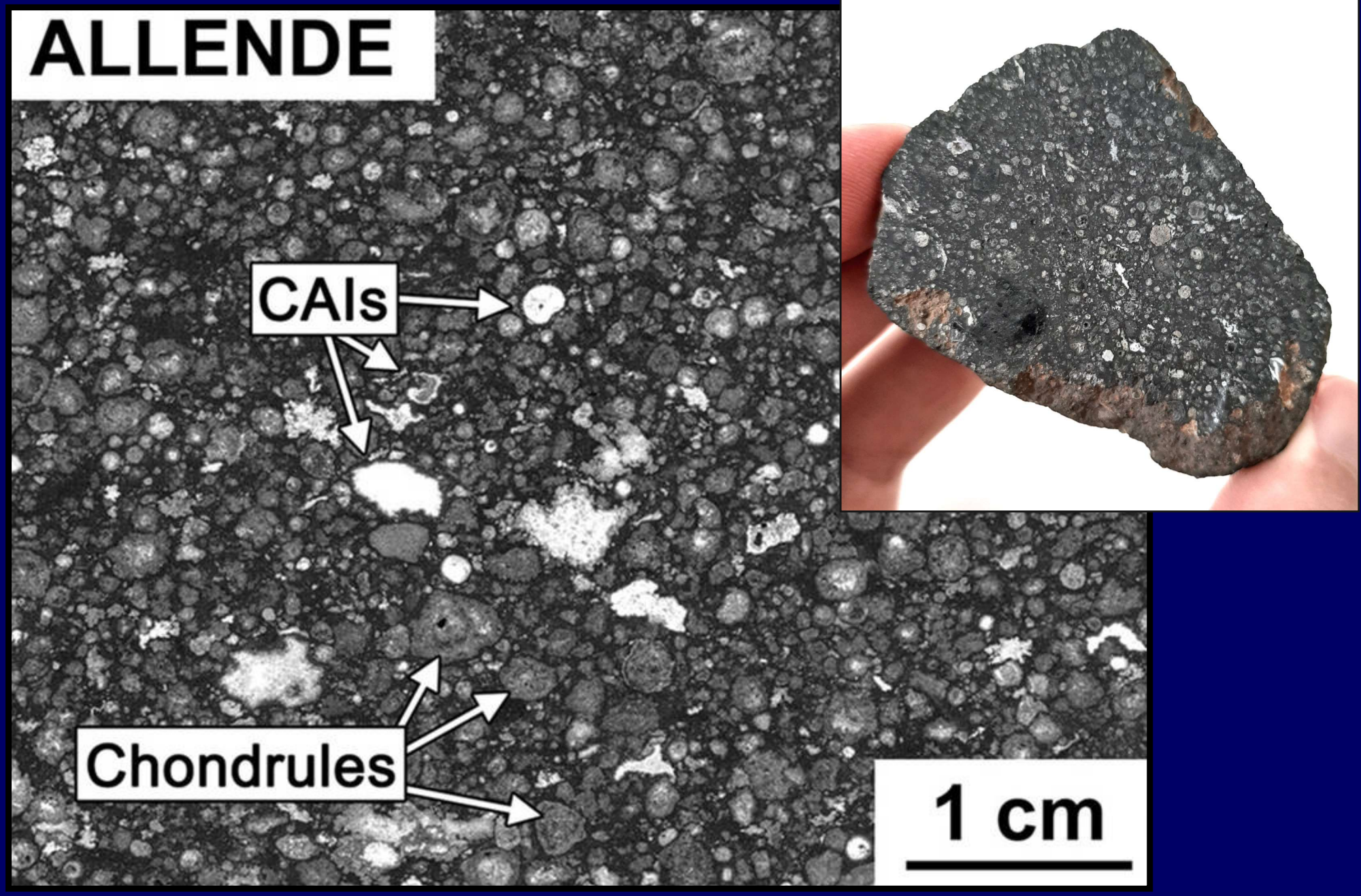
Formation of the planets



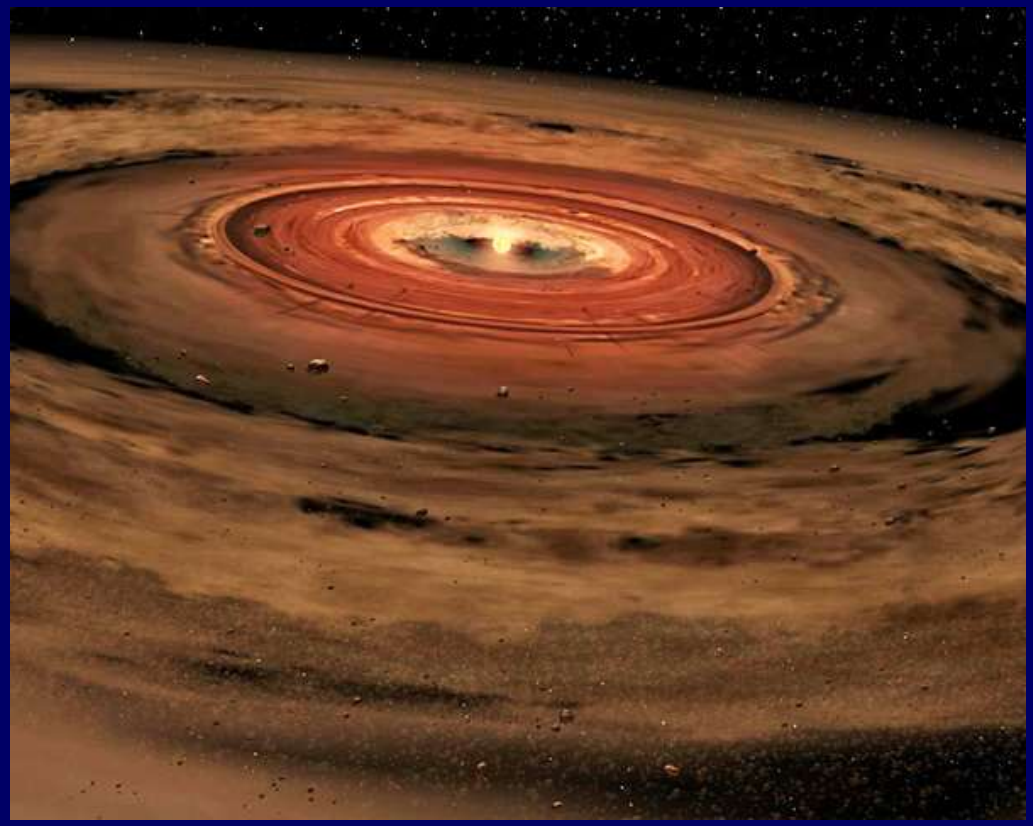
As the central part of the solar nebula contracted, the temperature rose

Early stages, T>1200 - 1300 K at Earth distances from the star → vaporization of most of the pre-solar dust

The disc: condensation



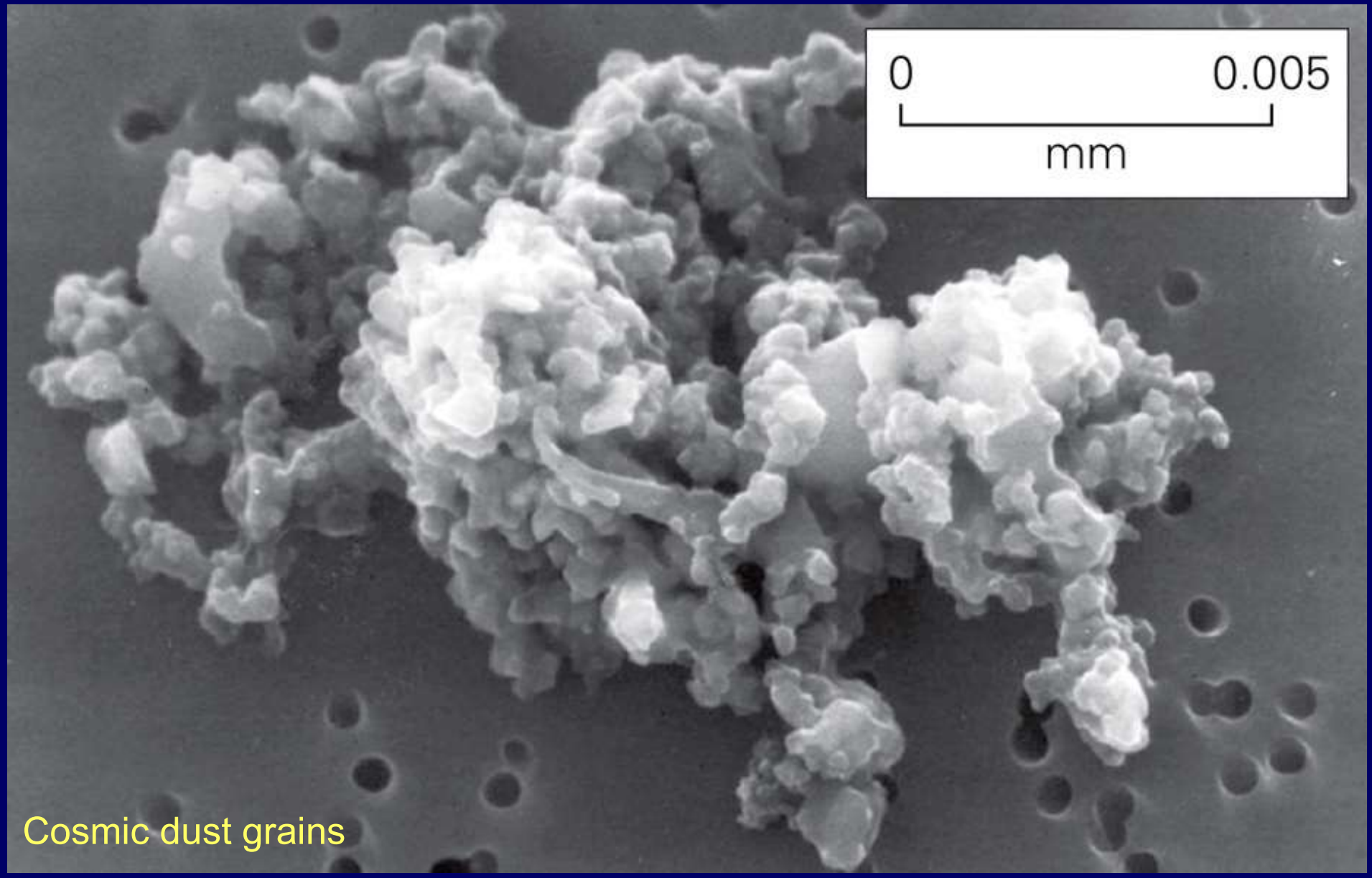
Stages in the formation of the planets



- Starting from tiny dust particles, larger and larger objects formed to create the asteroids and terrestrial planets

- This growth process is generally divided into three stages:
  - aggregation of micron-sized dust to 1- to 10-km-diameter planetesimals through coagulation, accretion and gravitational attraction
  - runaway growth of the largest planetesimals to form planetary embryos
  - aggregation of embryos to form the terrestrial planets

Coagulation



Cosmic dust grains

Accretion



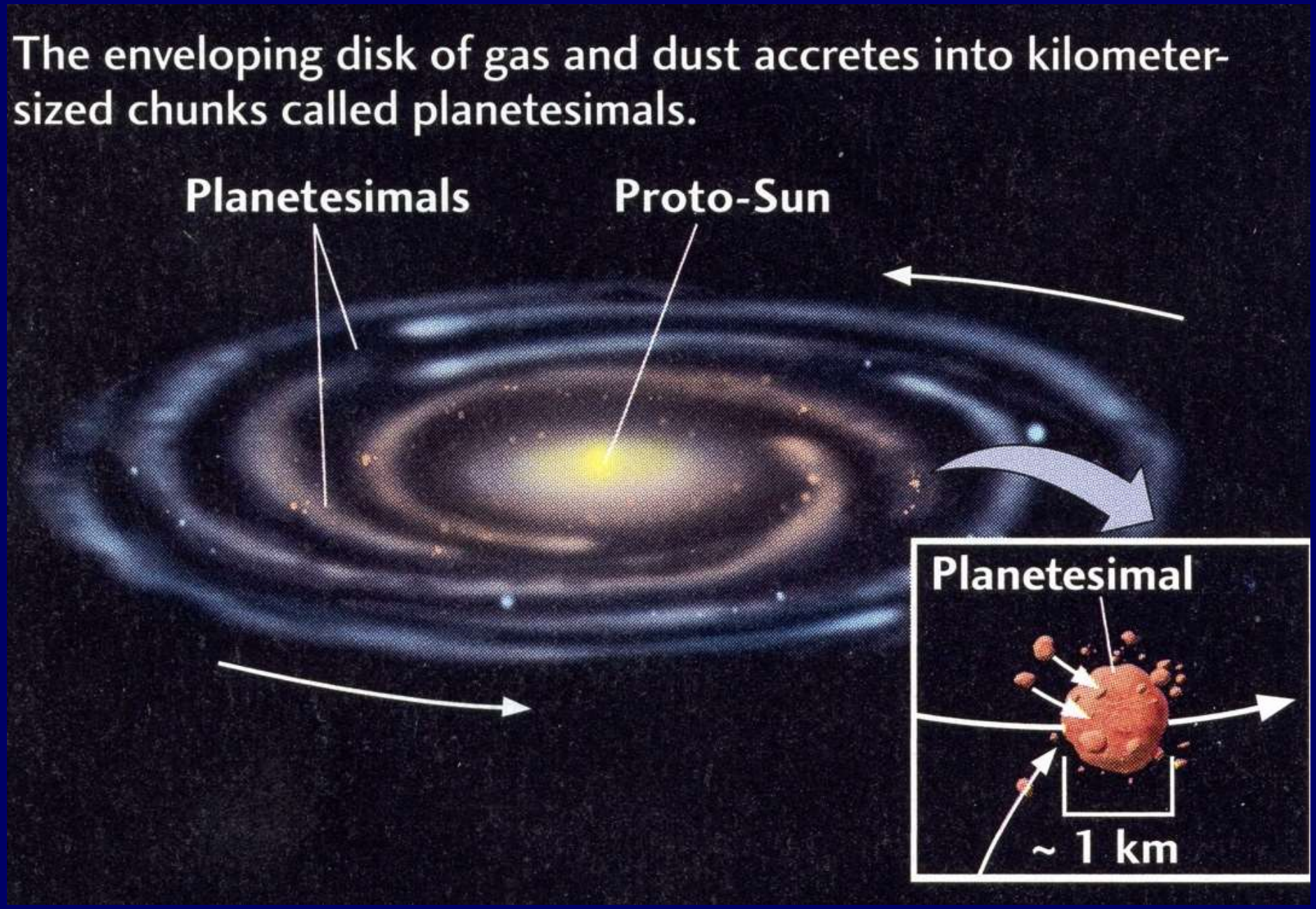
Dust accreted into grainy masses, similar to this meteorite fragment

Towards planetesimals

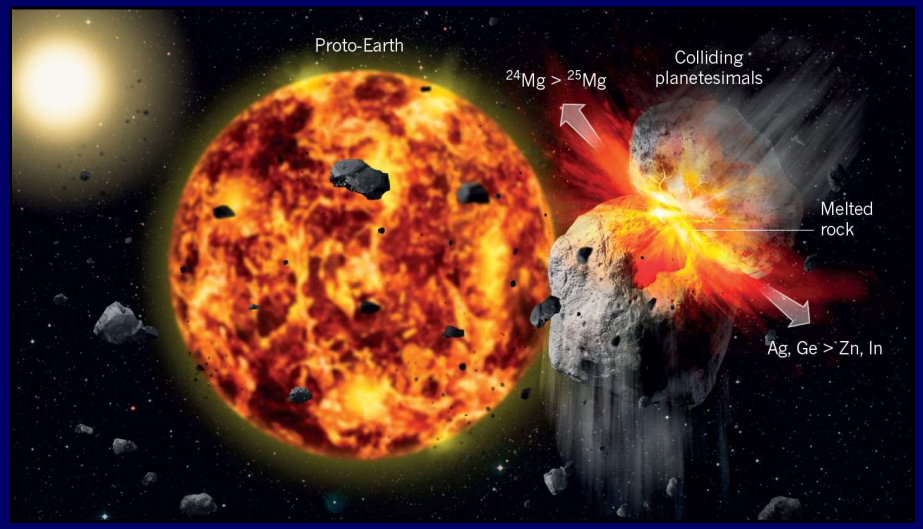


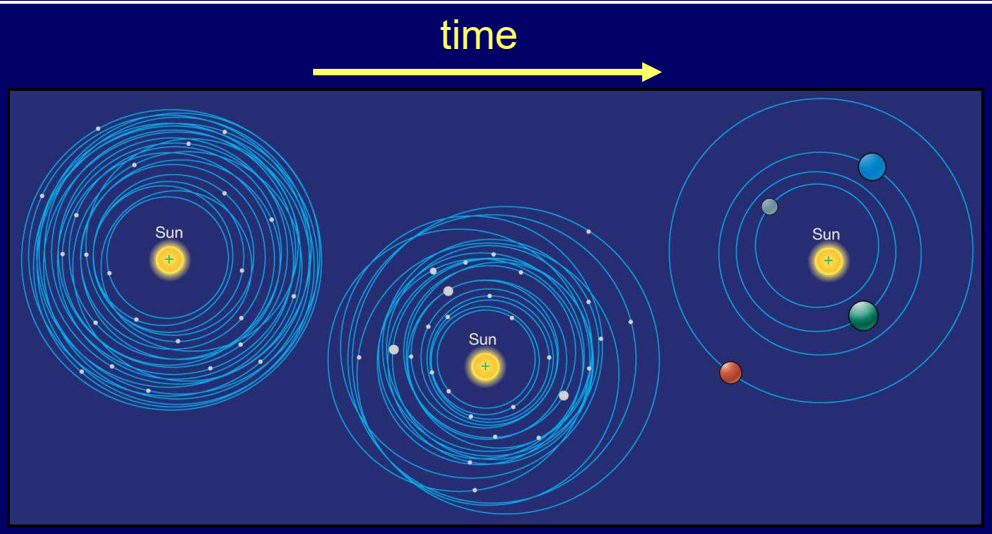
A ‘streaming instability’ (illustrated by a murmuration of starlings)

Planetesimals form

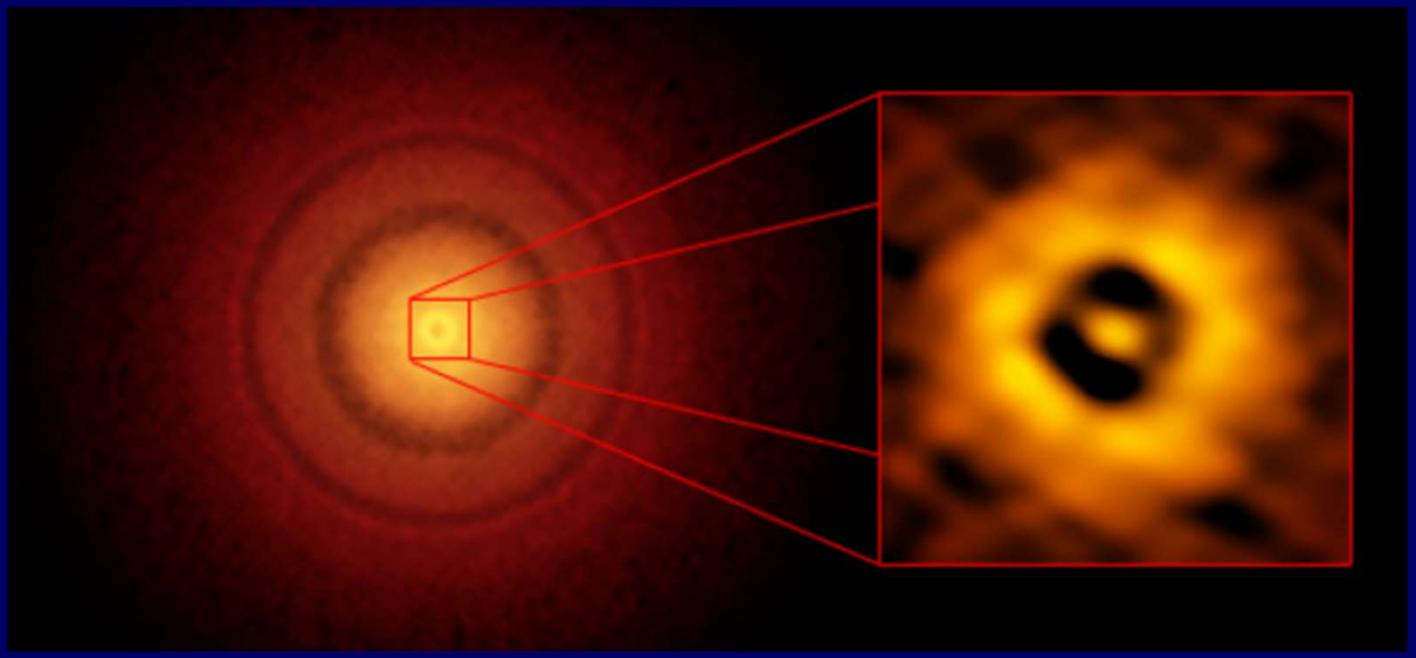


Planetary embryos form





Computer simulation of planetary accretion in the inner Solar System



ALMA image of the planet-forming disk around the young, Sun-like star TW Hydrae

Source of Heat for Melting



Formation of a magma ocean

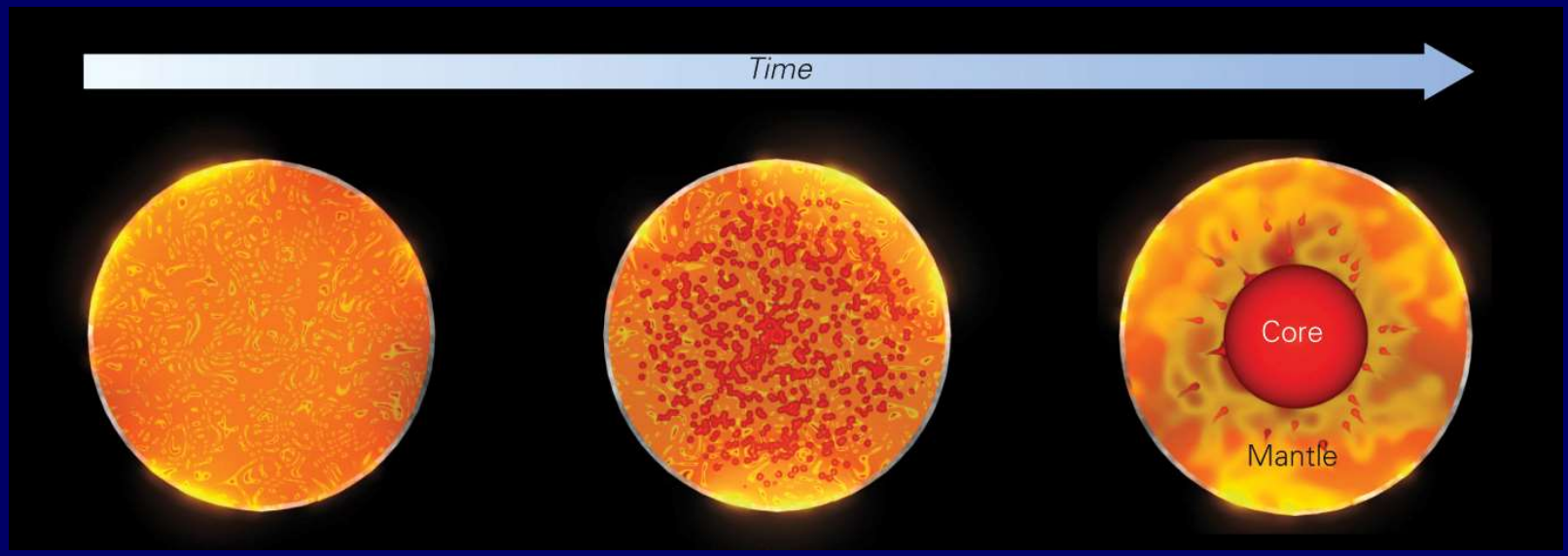
Heat released by impacts

- kinetic energy of impact is converted to heat
- complicated by size of object and assembly time

Heat created by decay of radioactive elements

- heat created by now extinct short-lived radionuclides
- heat from long-lived radionuclides

Gravitational Differentiation of the Earth's Interior



Early on, the Earth was fairly homogeneous inside

When the temperature got hot enough, iron began to melt

The iron accumulated at the center of the planet to form a metallic core

Melting resulted in the formation of magma ocean, separation of metals and molten silicates and differentiation of the Earth's interior into a mantle and core

Key characteristics of metal and silicate

1. The high density of metal relative to silicate means the metal would sink through the silicate under the force of gravity
2. Density separation is facilitated because metal and silicate are immiscible

Immiscibility and density contrasts lead to discrete layer with a sharp boundary between them

Time scales of Planetary Formation and Differentiation

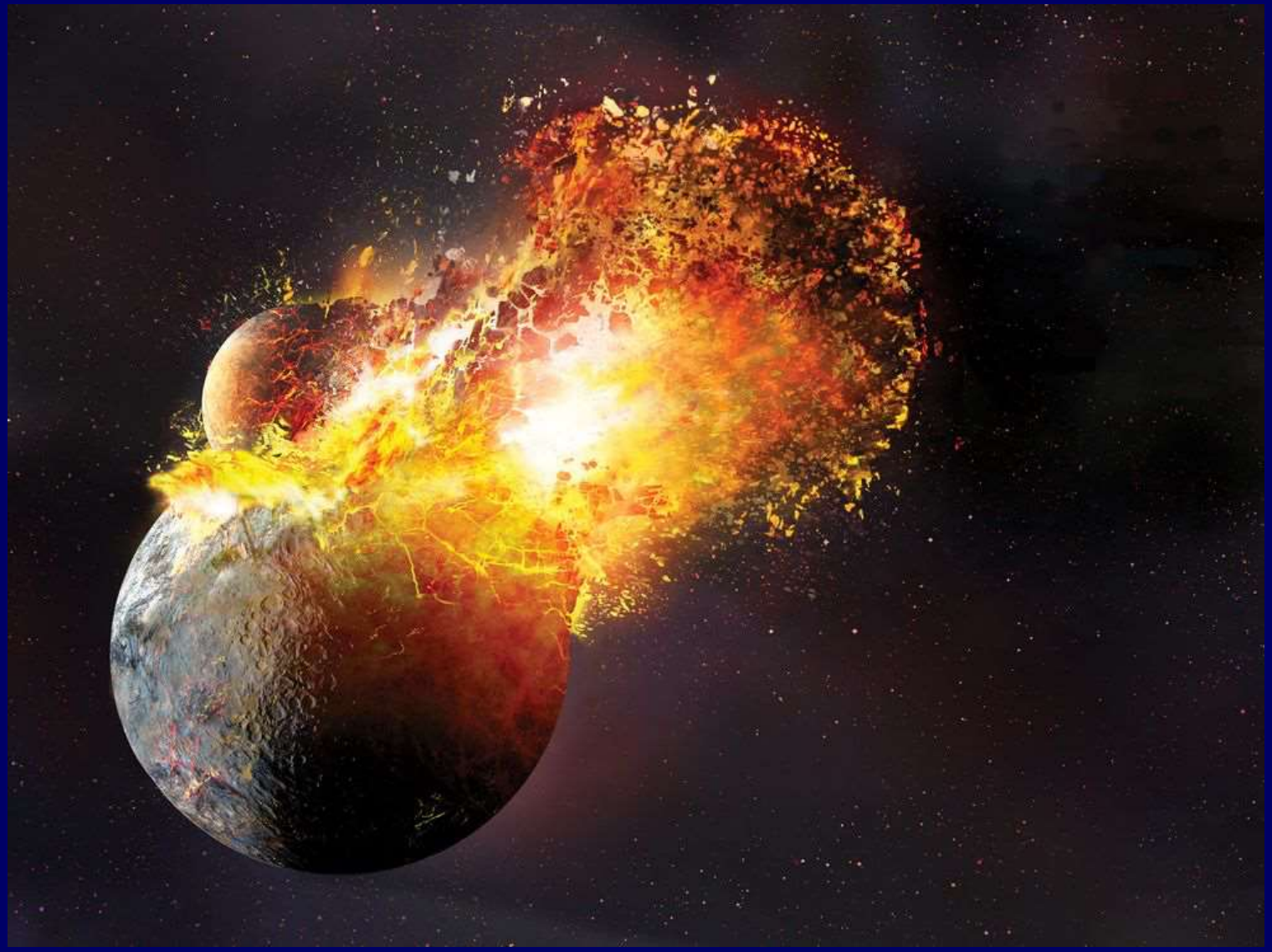


t_0 = age of solar system
= ~4567 Ma

Earth 63% formed by 11 myr after t_0 , and ~90% formed by ~ 30 myr after t_0
Core formation took place in the first 30 Ma of Earth history

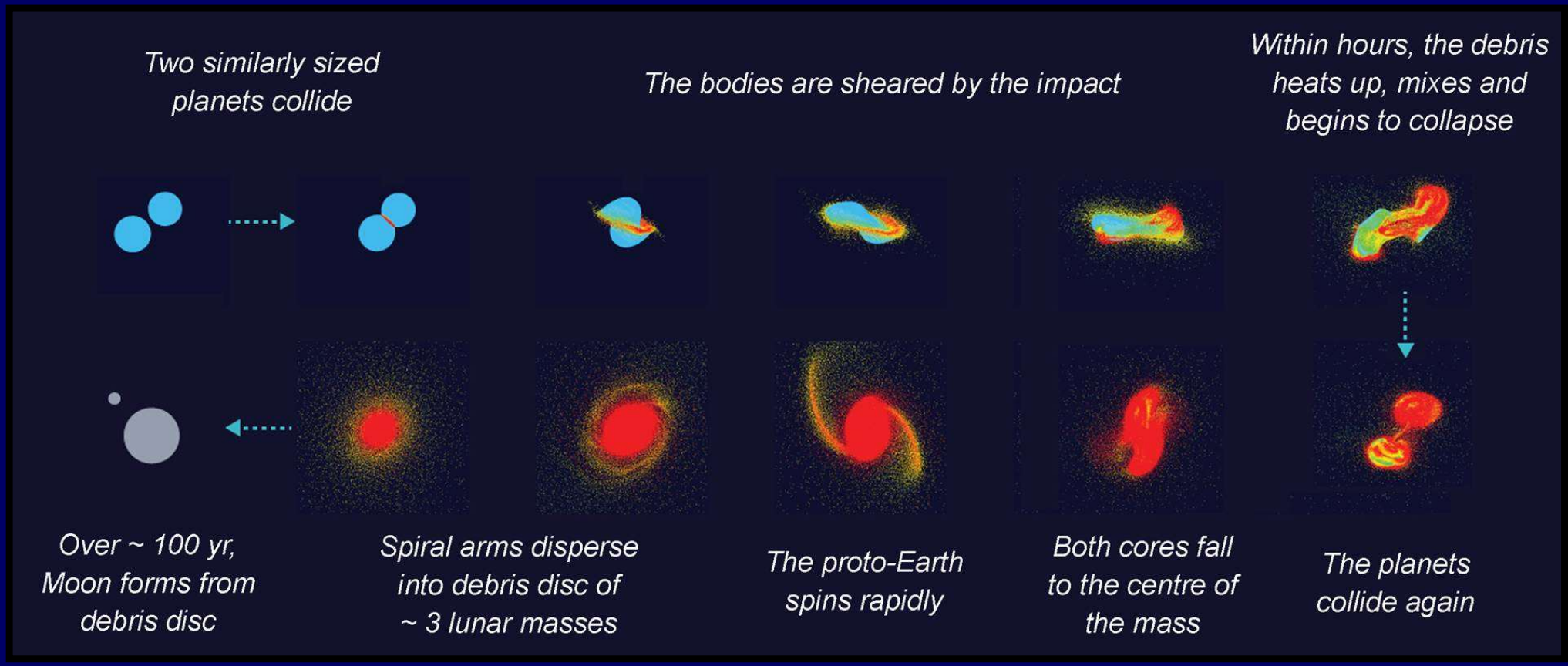
From the perspective of the age of the solar system (4567 Ma), planetary formation and differentiation, powered by impacts and extinct radionuclides, occurred within the first 1% of solar system history

The Formation of the Moon



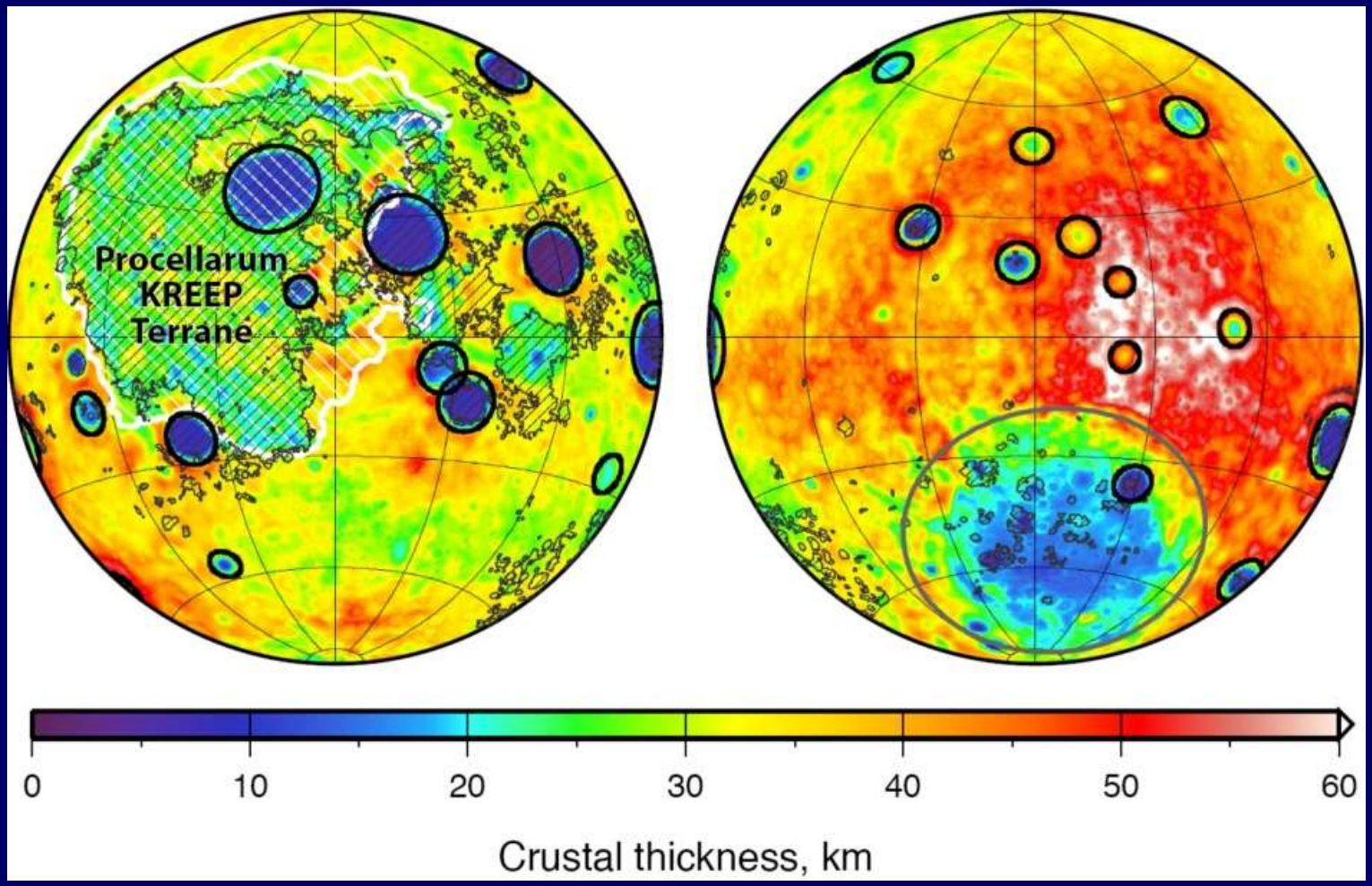
The Moon formed ~4500 Ma (before the oldest lunar rocks, 4460 Ma) as the result of a ‘giant impact’

The Giant Impact Hypothesis

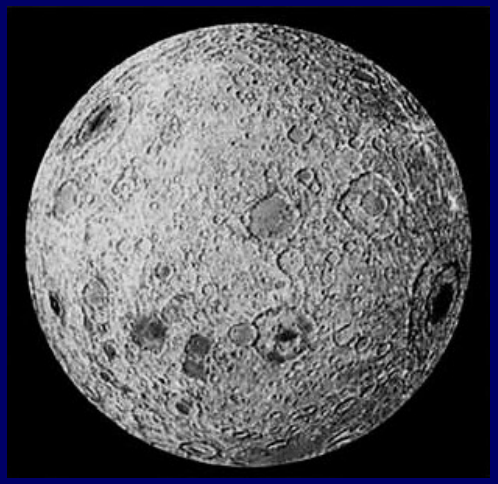


Late in the Earth's accretionary history, a protoplanet (Theia) collided with the Earth and vaporized the mantles of both bodies; some of this material was retained in Earth orbit to form a circumterrestrial disc from which the Moon accreted shortly afterward

Mysteries of the Far Side of the Moon





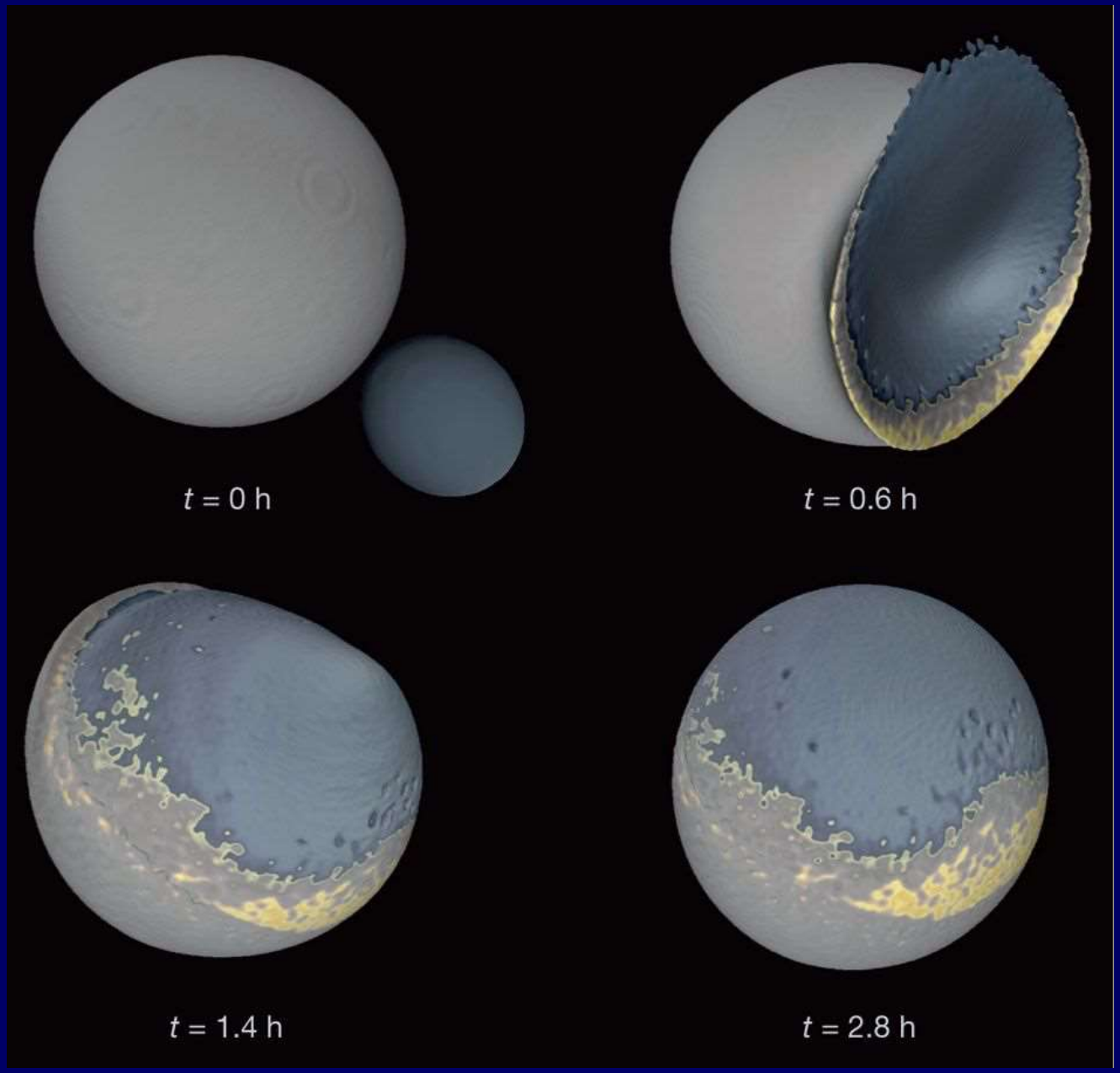


Observation

2012 NASA's GRAIL mission showed key differences between the near and far side of the Moon :

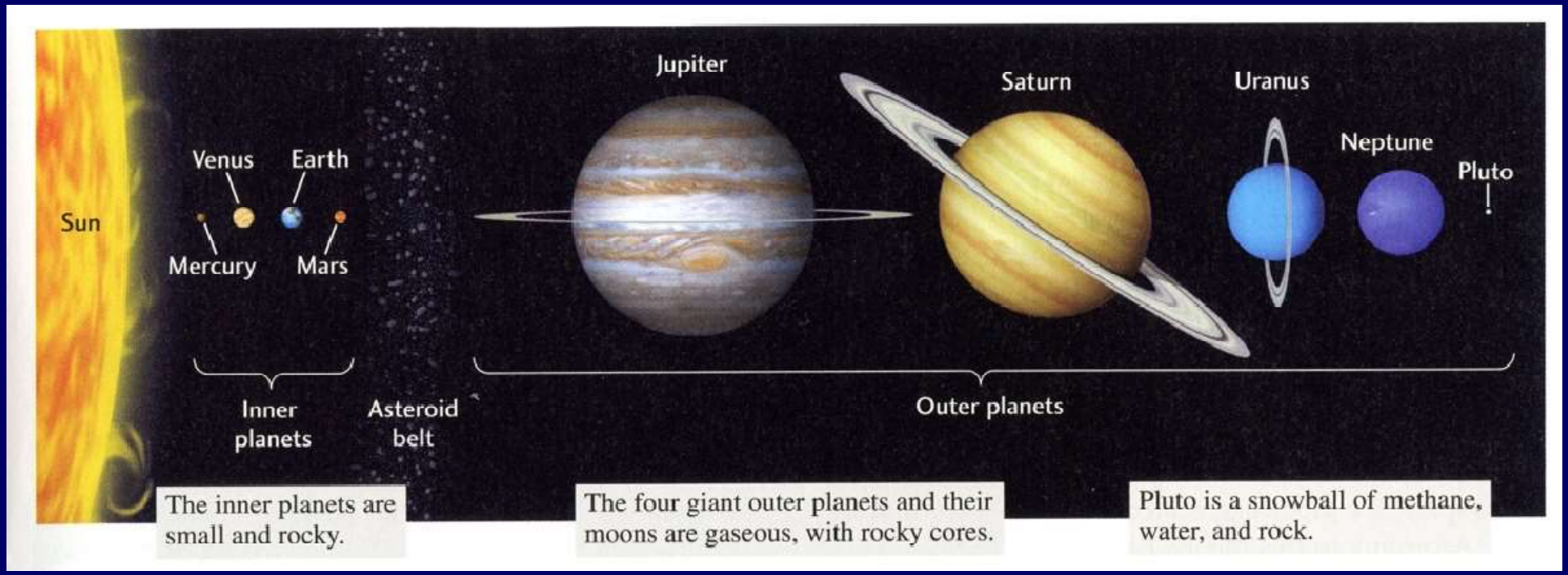
- Much thicker crust (up to 60 km in highlands)
- Absence of maria

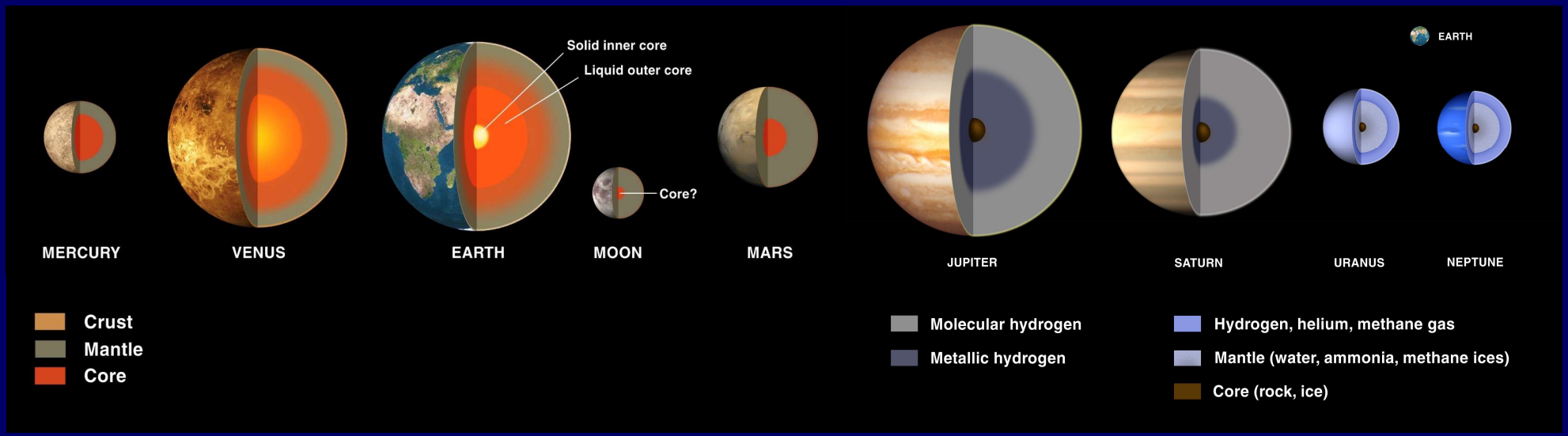
The Big Splat



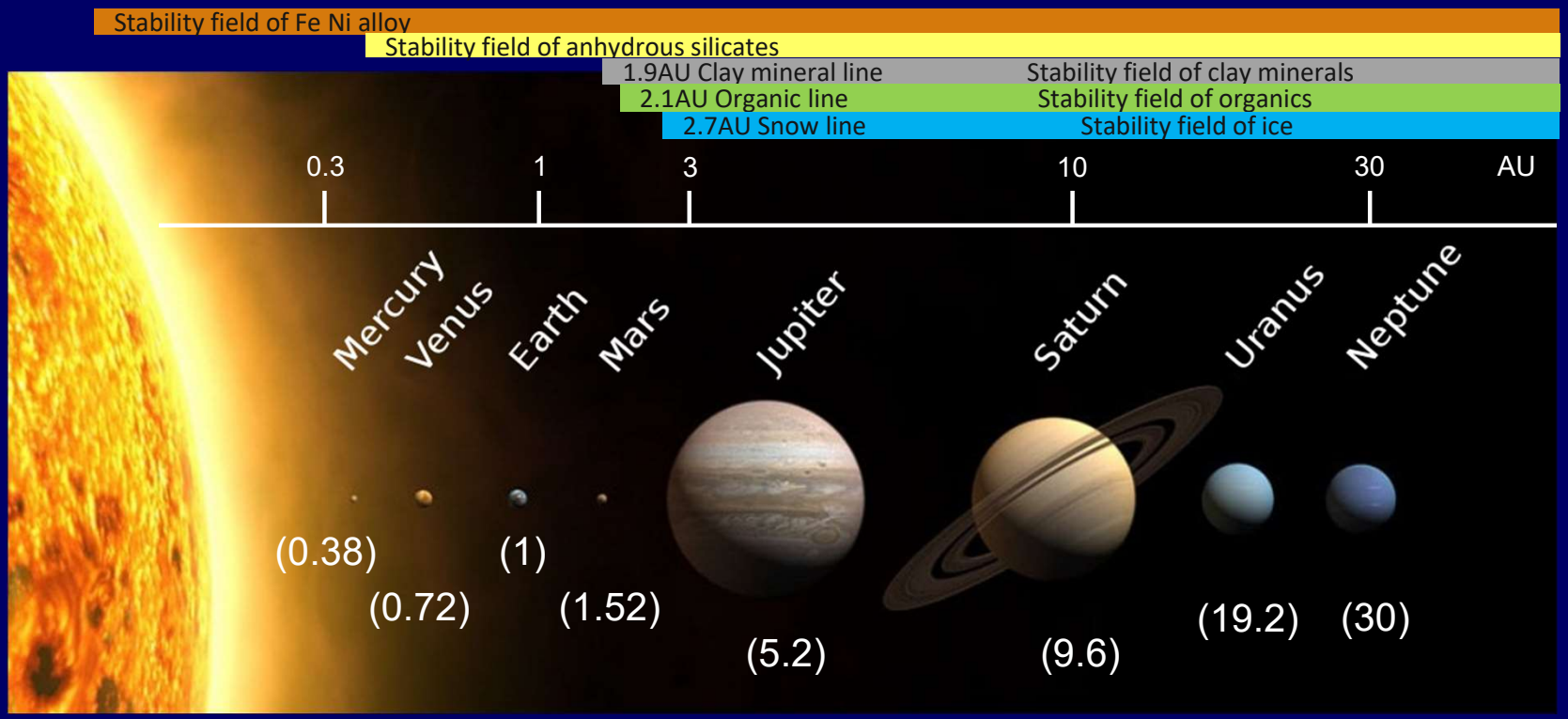
Slow impact (2 km/s) smaller Moon splatted/smeared across far side of Moon

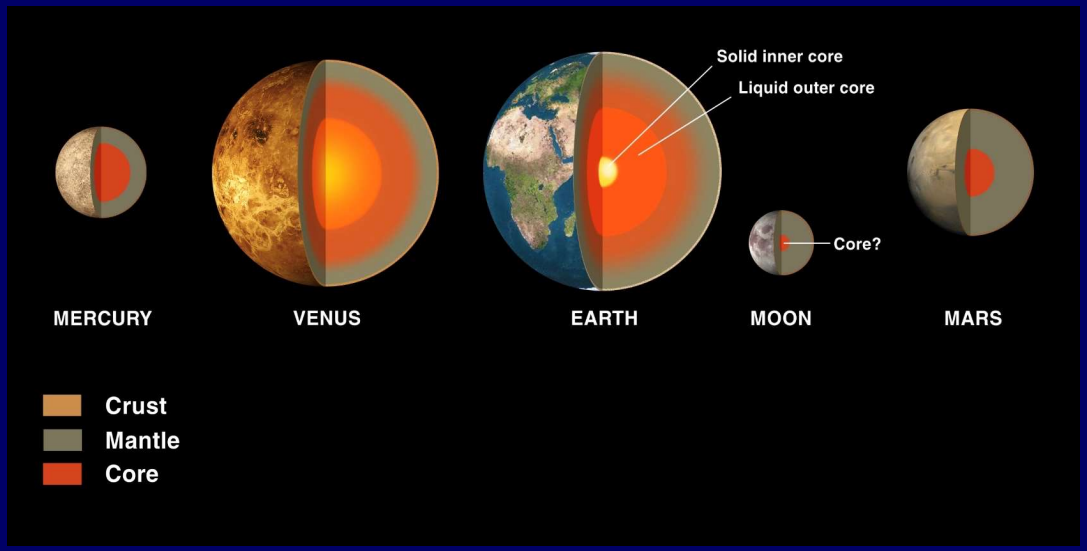
Summary: Comparison between the planets

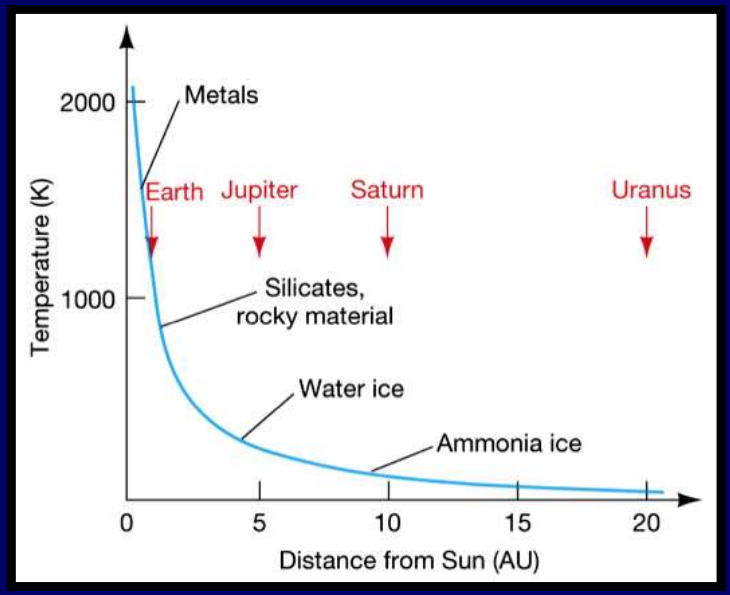




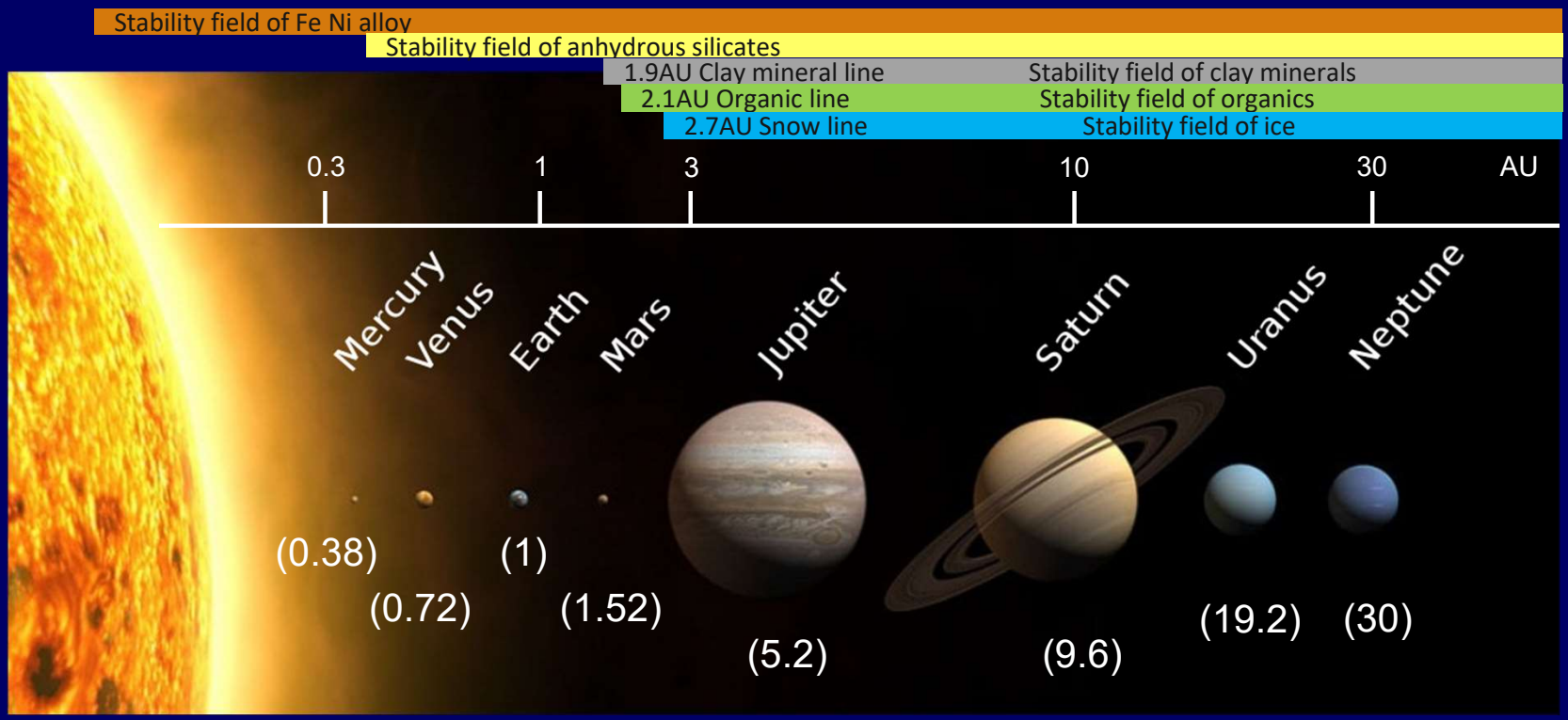
Summary: The Inner Planets

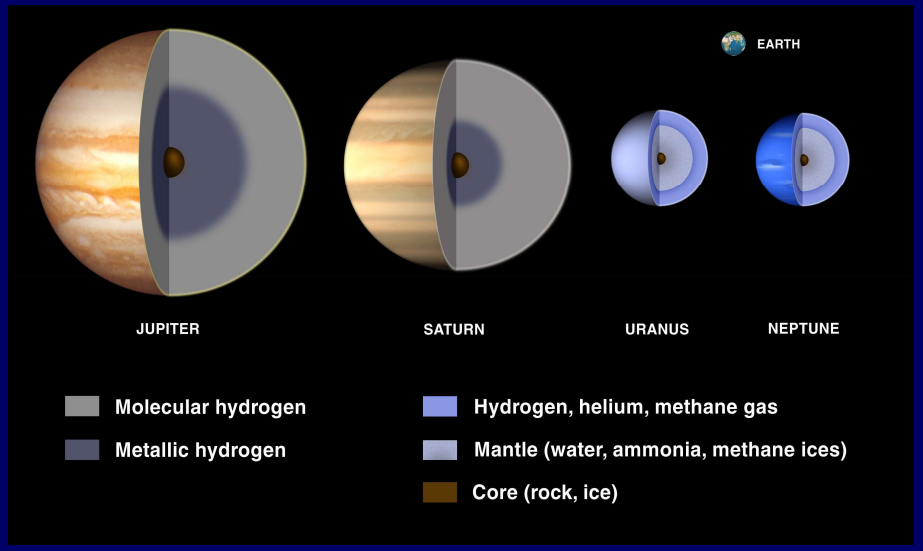


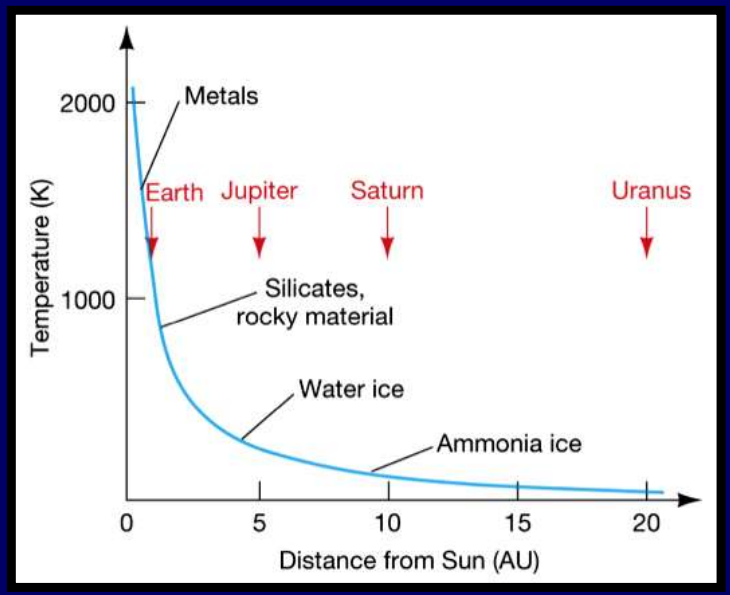




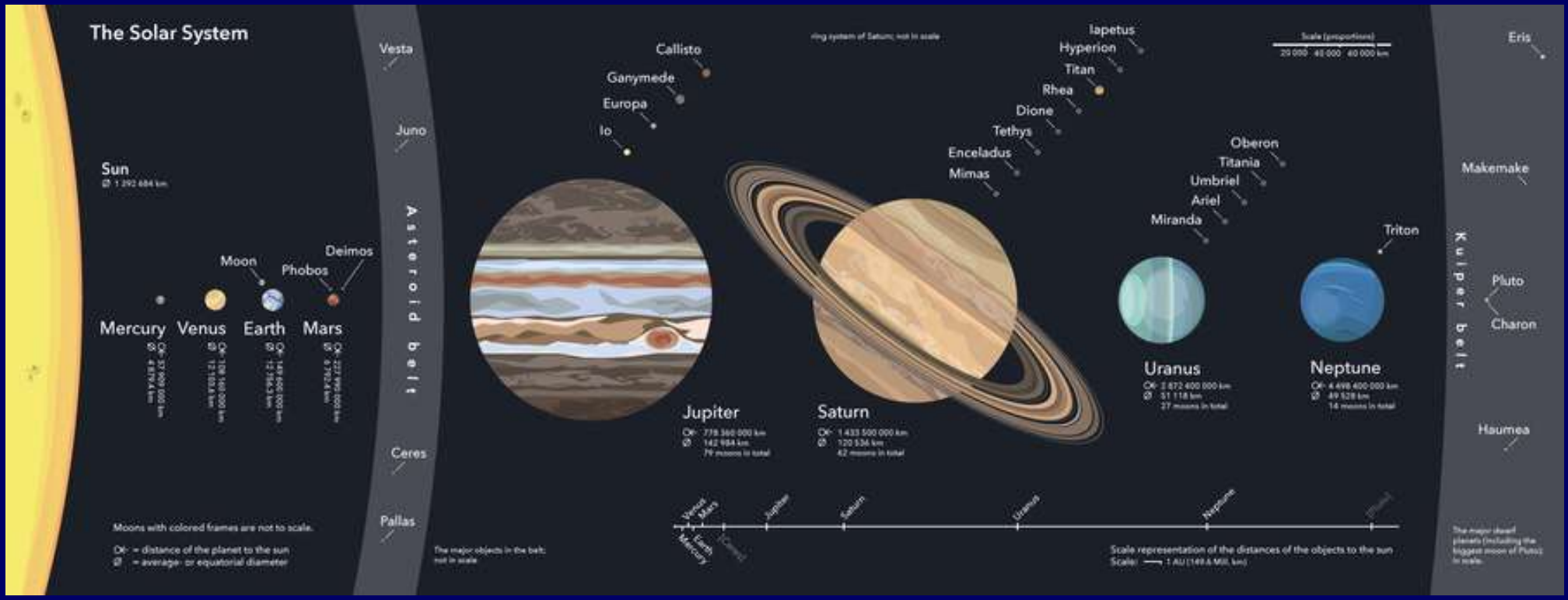
Summary: The Outer Planets







Summary: Timeline for the early Solar System



Formation of Universe: 13797 Ma

Formation of Milky Way: 13200 Ma

Formation of Solar System: 4567 Ma

Now: 0 Ma

MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

Week 3

The Hadean Earth and its Environment

4567-4031 Ma

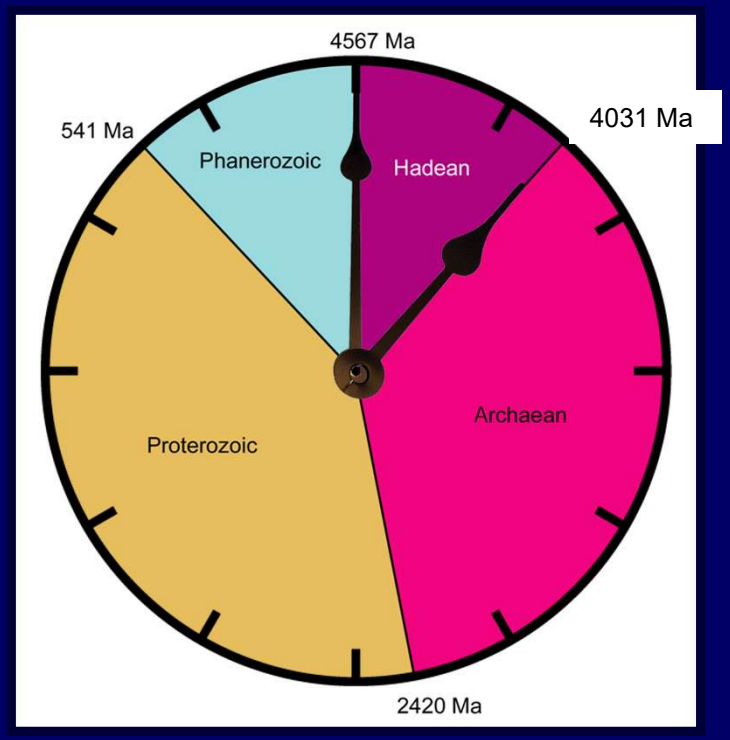
age of the solar system

oldest rocks



By the end of this class you will be able to compare the early Earth in the Hadean to the present day in terms of tectonic style and environmental conditions

The Hadean Earth: the ‘unseen place’

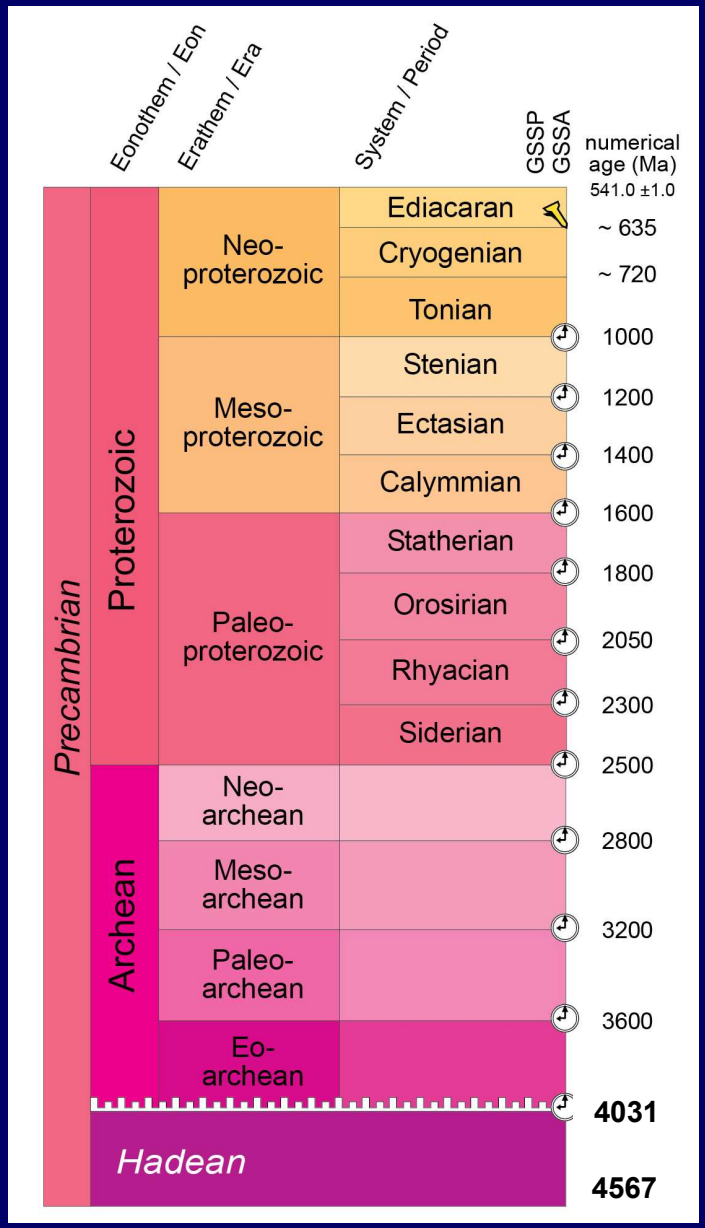




4031 Earth’s oldest rocks (Acasta Gneiss)

4404 Earth’s oldest crustal material (detrital zircons)

4567 Formation of the solar system



Note spelling: Archaean Palaeo

The Crust of the Early Earth

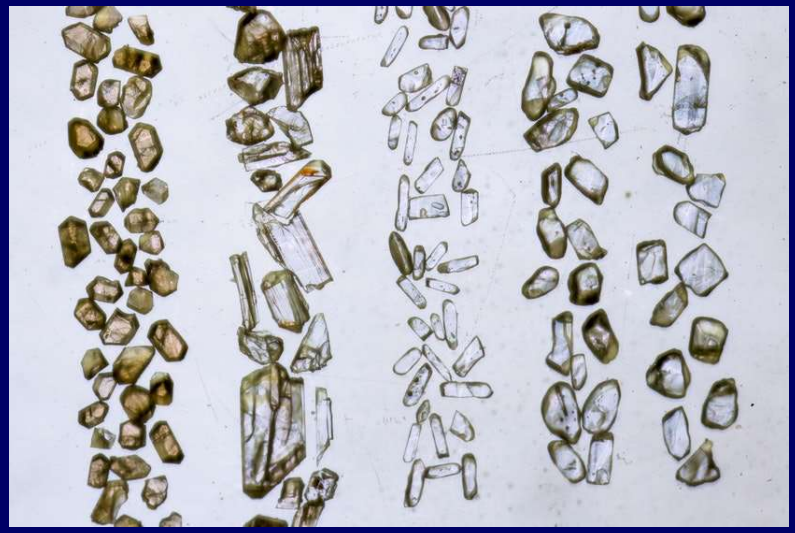


Jack Hills, Western Australia

No Hadean crust survives but Hadean detrital zircon grains contained within younger rocks do – almost all these have been found in Western Australia

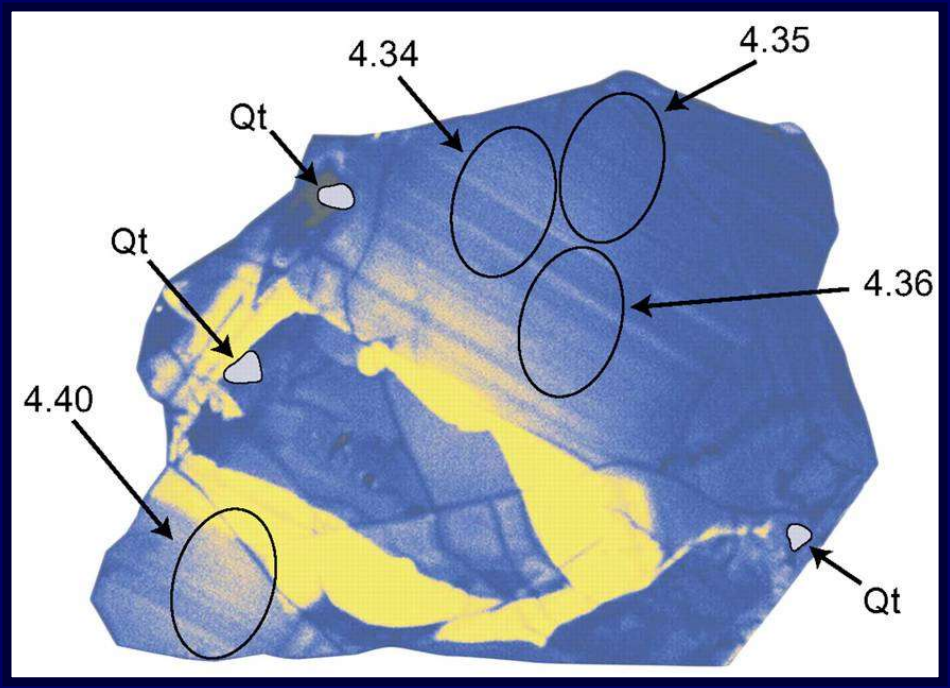
Zircon is a mineral commonly found in granite rocks





Zircon crystals found in sandstone at Jack Hills

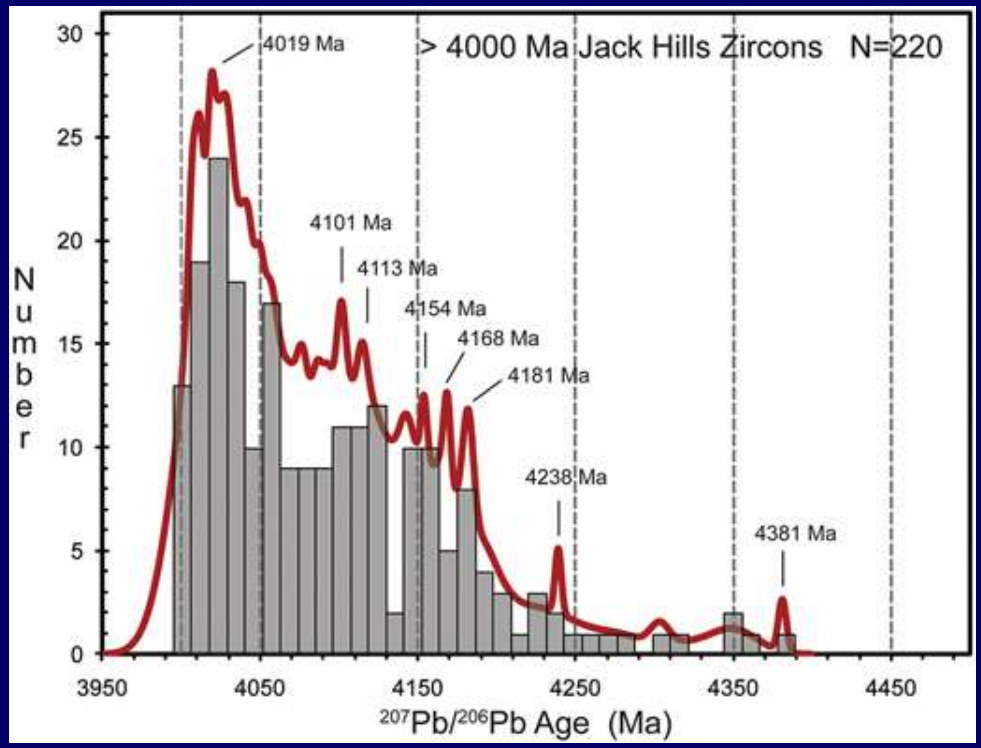
The Oldest Dated Crustal Material



The oldest dated zircon crystal; ages in Ga

- Polished surface of a zircon grain from Jack Hills
- The mineral zircon = $ZrSiO_4$
- Spots are ~ 50 microns across
- Ages are U-Pb dates in billions of years (Ga)

- Presence of quartz inclusions (Qt) indicates zircon sources were granite rocks
- Granite formed by melting pre-existing basalt crust in presence of liquid water → indicates magma ocean phase complete by 4.4 Ga
- Little chemical variation in composition of mafic crust for ~300myrs: crust very long-lived → implies modern plate tectonics not exist in Hadean ('stagnant lid'- single, fixed plate)



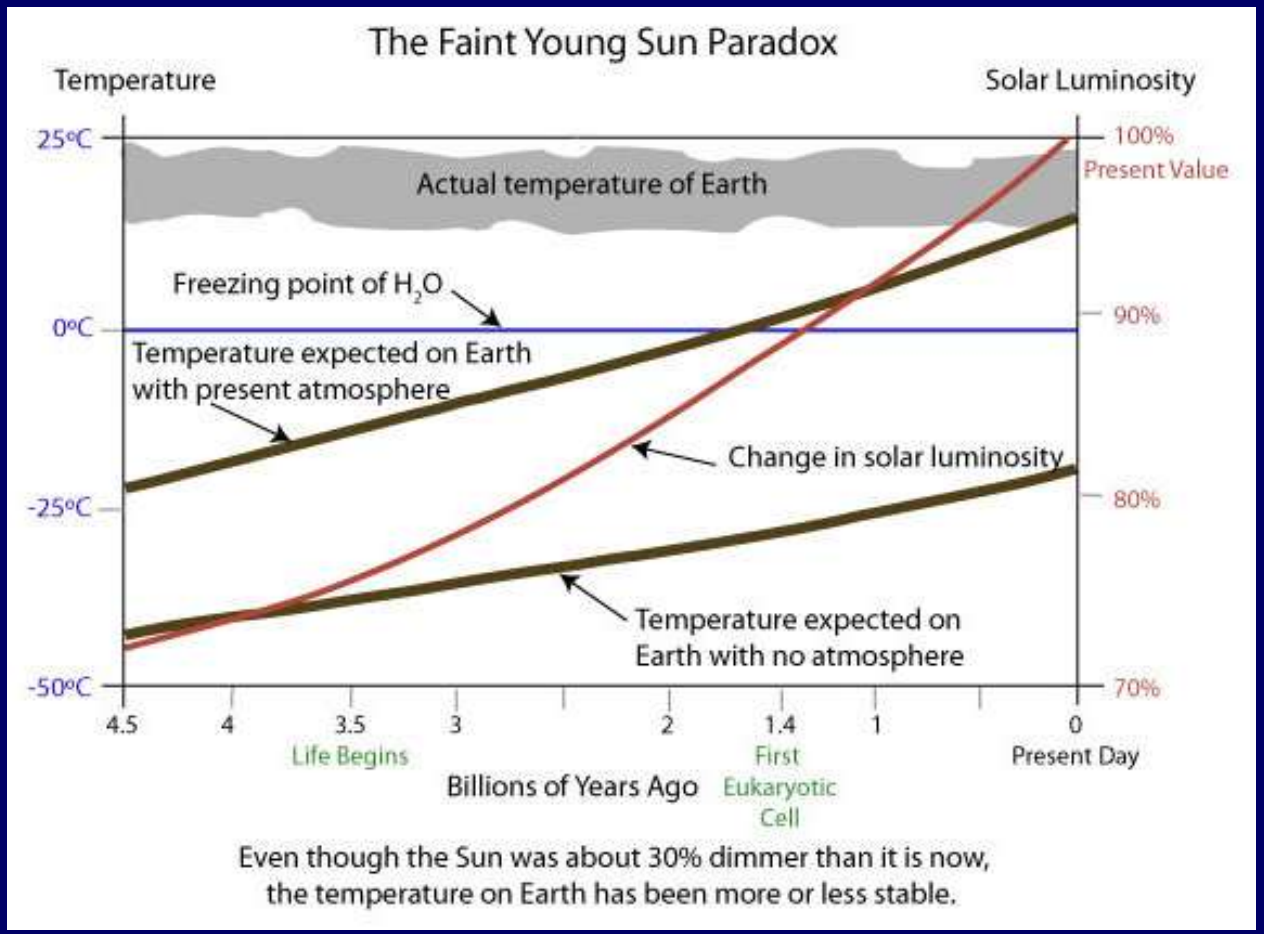
The distribution of zircon ages at Jack Hills

The Hadean Environment



Moon closer to Earth → Earth spinning faster → days shorter (2-3hr)

What was the nature of the Earth's atmosphere and temperature?



Estimates of average temperature of Earth

Sun ~70% as luminous as today

- If no atmosphere: -47°C
- If current atmosphere: -25°C

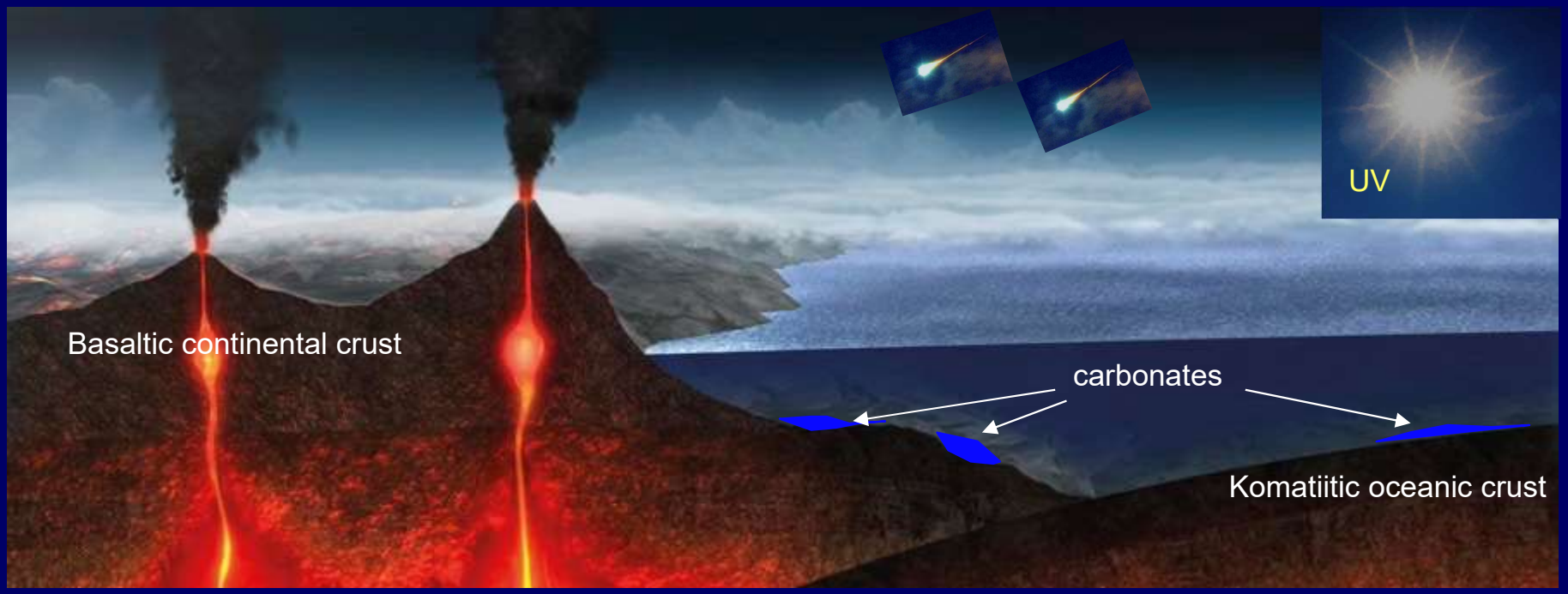
But isotopic composition of zircons suggests weathering → liquid water!

Liquid water → presence of atmosphere with high concentrations of greenhouse gases

Evolution of the Hadean Atmosphere

Following Moon impact:

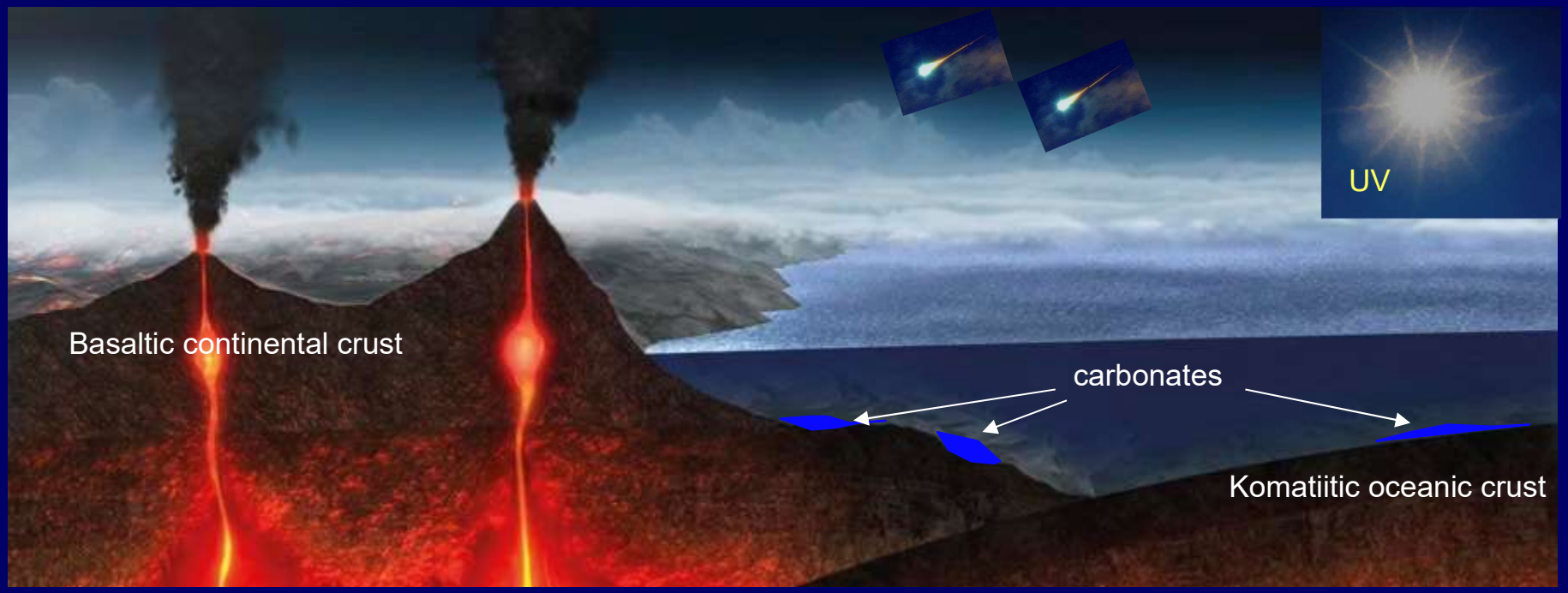
- atmosphere dense, deep and very toxic (CO_2-CO-H_2O-H_2)
- surface too hot for stable, liquid water

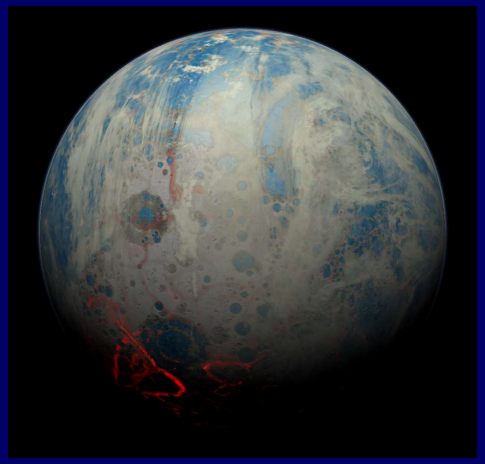


After ~10 million years Earth cooled sufficiently for water to condense from atmosphere

- crust covered with warm, carbonated ocean
- dense CO_2-rich atmosphere
- 'Goldilocks zone' maintained by greatly elevated levels of CO_2 and methane from volcanic out-gassing or impact-vaporized materials

Evolution of the Hadean Atmosphere





Artistic illustration of the early Earth

Removal of CO_2 from atmosphere to form carbonates in oceans → rapid cooling and oceans cold and icy

Multiple atmospheres in history of early Earth
Impacts → cycles of warming and cooling and ephemeral oceans

The Hadean Landscape: Implications for life



A Dantean Hell…..

Early impacts caused hostile conditions for life

- 3-7 impact events large enough to vaporise oceans

Long-term beneficial effects - stabilise surface temperatures and deliver key elements (H_2O and sulphur)

Persistent niches for prebiotic chemistry → life by 4400my?

- graphite inclusions in 4100 Ma zircon with carbon isotope signature consistent with biological origin

Impacts following the initial accretion of Earth and relocation of Giant Planets were probably sporadic rather than continuous



…. or a barren world dominated by a broad ocean and rare islands

Intended Learning Outcome Revision

Now we have completed the class, you should be able to:

- describe the formation of planets from a solar nebular
- state the age of our own solar system
- describe the processes and time scales over which the Earth's interior became gravitationally differentiated
- state when the Moon formed and compare hypotheses for its origin
- identify key differences between the planets of the solar system
- compare the early Earth in the Hadean to the present day in terms of tectonic style and environmental conditions

Follow up……

Watch the videos in the **Exploration of the Modern Earth** folder and make your notes on the internal structure of the modern Earth

Revise using the study guide on Blackboard

Next week……

What the Earth is made of

- Rocks
- Minerals